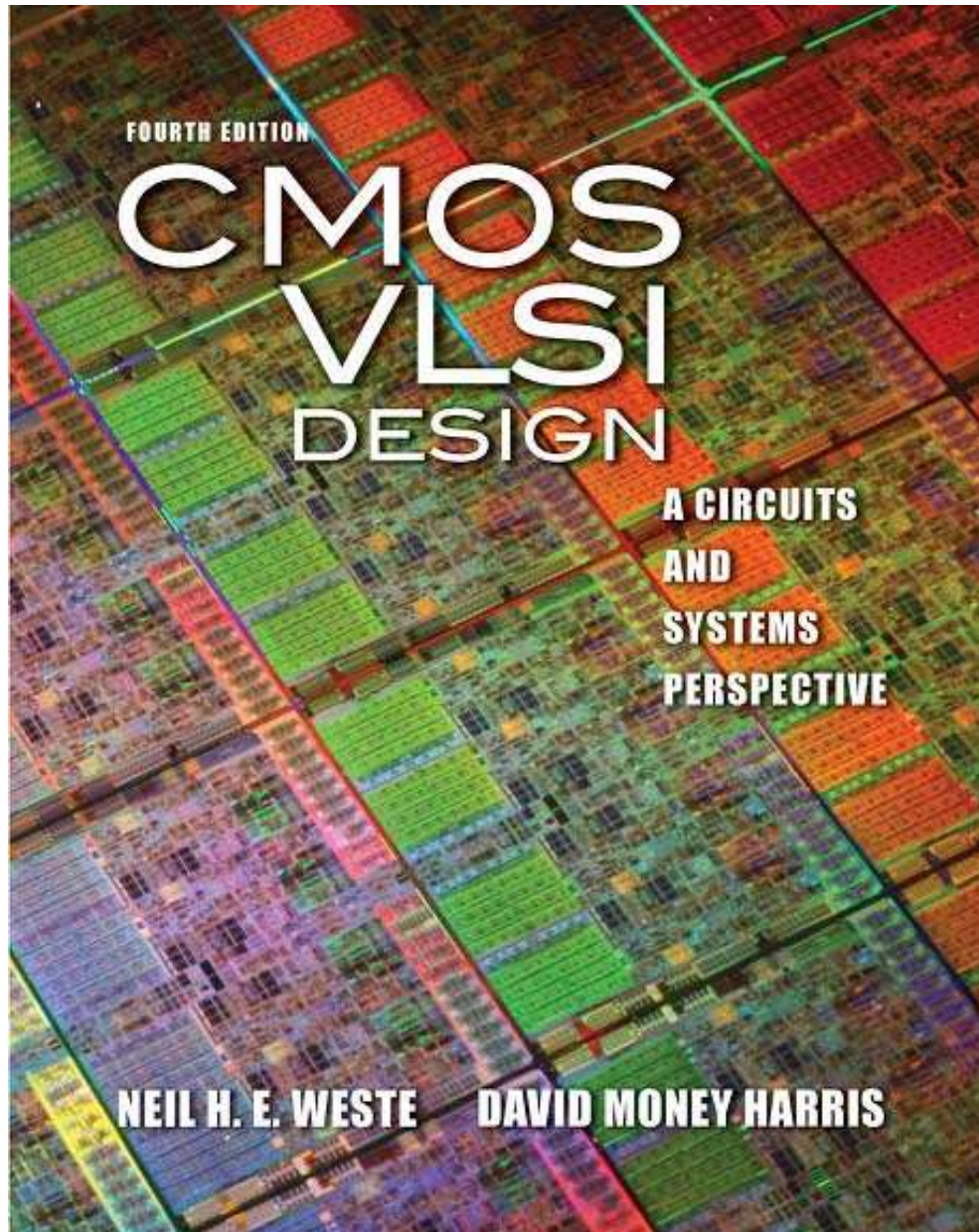




Chapter 2: MOS Transistor Theory

1. MOS transistor operation
2. I-V characteristics
3. C-V characteristics
4. DC transfer characteristics

1. MOS Transistor Operation

- ❑ So far, we have treated transistors as ideal switches
- ❑ An ON transistor passes a finite amount of current
 - Depends on terminal voltages
 - Derive current-voltage (I-V) relationships
- ❑ Transistor gate, source, drain all have capacitance
 - $I = C (\Delta V / \Delta t) \rightarrow \Delta t = (C / I) \Delta V$
 - Capacitance and current determine speed

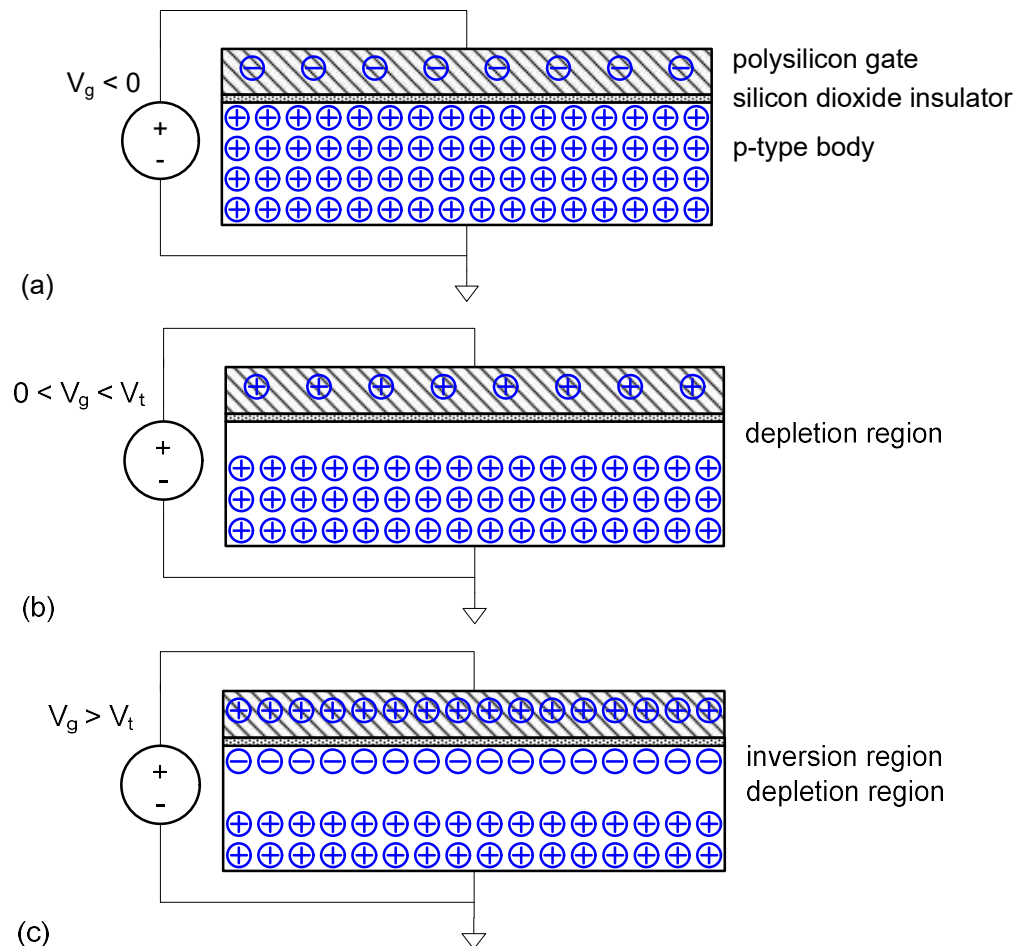


Electrical Property of MOS Devices

- ■ **Necessary to understand the basic electrical properties of the MOS transistor (geometry => electrical), e.g., delay/power**
 - Ensure that the circuits are robust
 - Create working layouts
 - Predict delays and power consumption
- ■ **As technology advances and circuit dimensions scale down, Electrical effects become more important**
 - Secondary/non--ideal effects (next lecture)

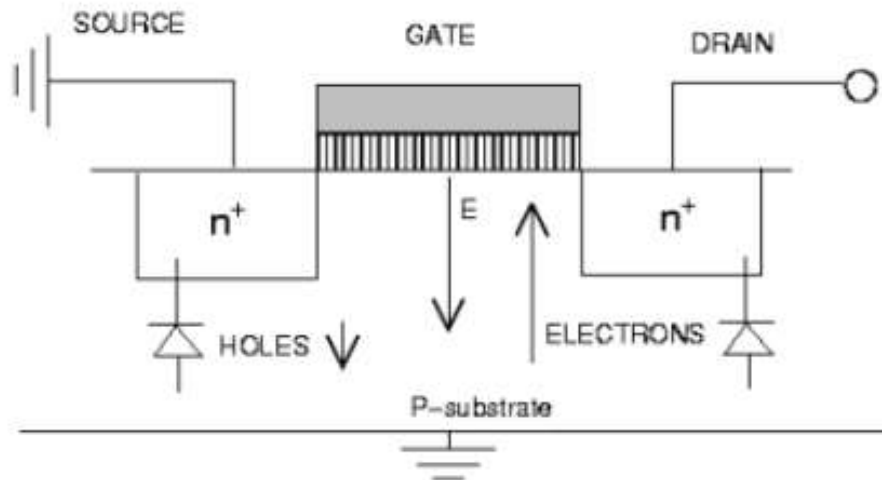
MOS Capacitor

- ❑ Gate and body form MOS capacitor
- ❑ Operating modes
 - Accumulation
 - Depletion
 - Inversion



MOS Capacitor

The nMOS transistor

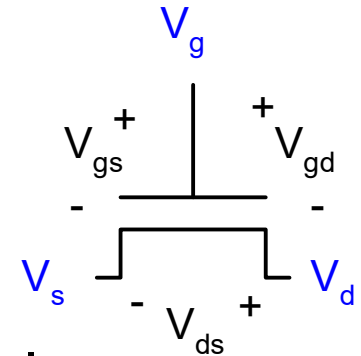


Moderately doped p- type substrate (or well) in which two heavily doped n+ regions, the Source and Drain are diffused

- **Gate is insulated from substrate by thin oxide**
 - Resistance of oxide is $> 10^{12} \Omega$, so current ~ 0
- **Two types of nMOS transistor**
 - Enhancement mode: non conducting when gate voltage $V_{gs} = V_{sb}$ (source voltage) (**normally used**)
 - Depletion mode: conducting when $V_{gs} = V_{sb}$

Terminal Voltages

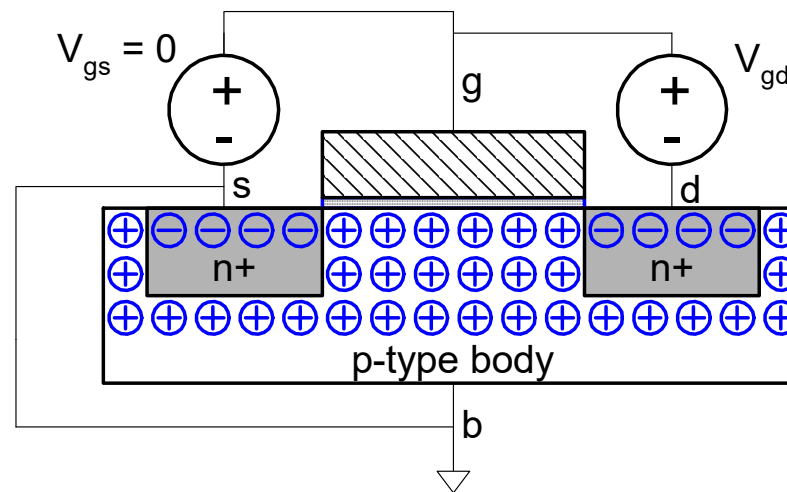
- ❑ Mode of operation depends on V_g , V_d , V_s
 - $V_{gs} = V_g - V_s$
 - $V_{gd} = V_g - V_d$
 - $V_{ds} = V_d - V_s = V_{gs} - V_{gd}$
- ❑ Source and drain are symmetric diffusion terminals
 - By convention, source is terminal at lower voltage
 - Hence $V_{ds} \geq 0$
- ❑ nMOS body is grounded. First assume source is 0 too.
- ❑ Three regions of operation
 - *Cutoff*
 - *Linear*
 - *Saturation*



nMOS

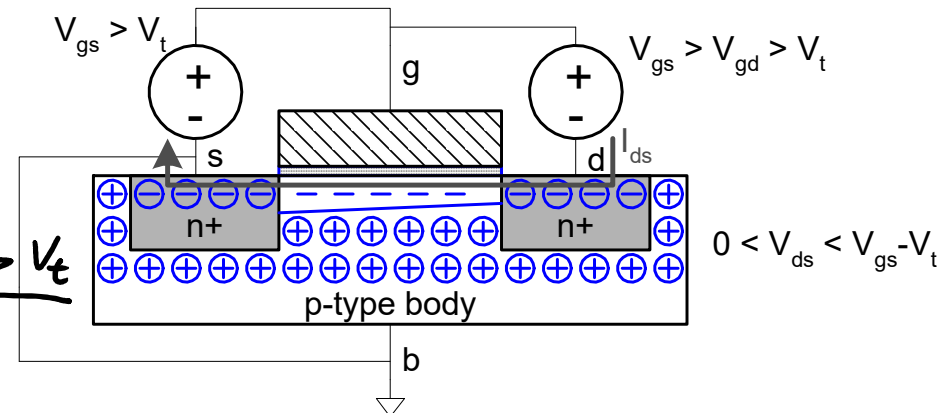
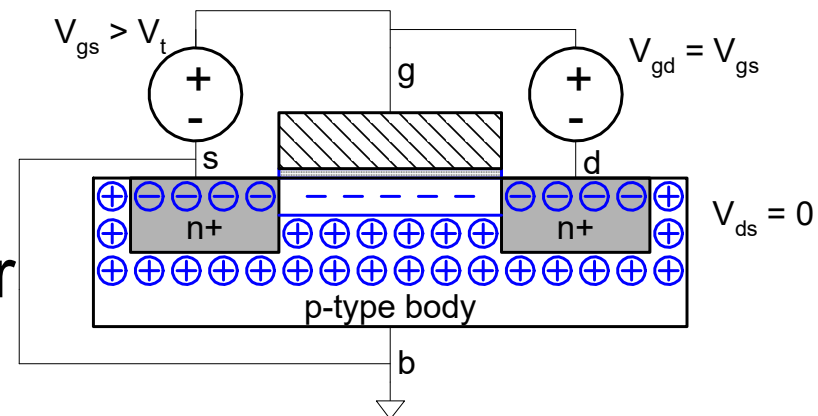
nMOS Cutoff

- ❑ No channel
- ❑ $I_{ds} \approx 0$



nMOS Linear

- ❑ Channel forms
- ❑ Current flows from d to s
 - e^- from s to d
- ❑ I_{ds} increases with V_{ds}
- ❑ Similar to linear resistor

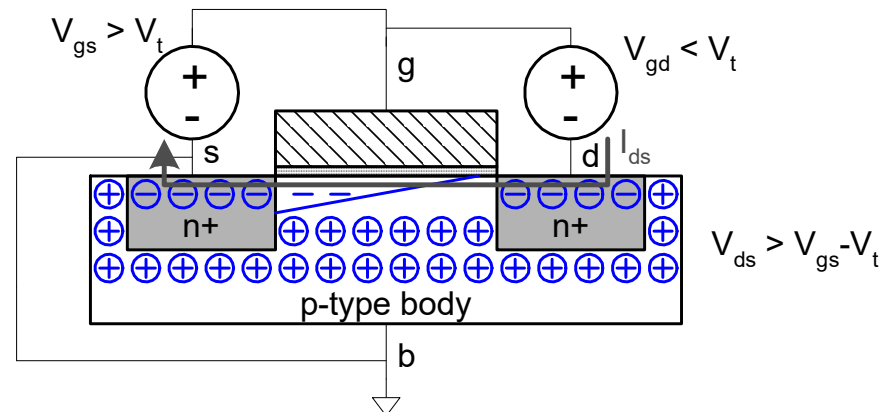


$$\begin{aligned}
 \textcircled{1} \quad & V_{ds} < V_{gs} - V_t \\
 & V_{ds} - V_{gs} < -V_t \\
 & V_d - V_s - V_g + V_s < -V_t \\
 & V_{dg} < -V_t \iff -V_{gd} < -V_t \iff \underline{V_{gd} > V_t} \\
 \textcircled{2} \quad & V_{ds} > 0 \iff V_{gs} - V_{gd} > 0 \iff \underline{V_{gs} > V_{gd}}
 \end{aligned}$$

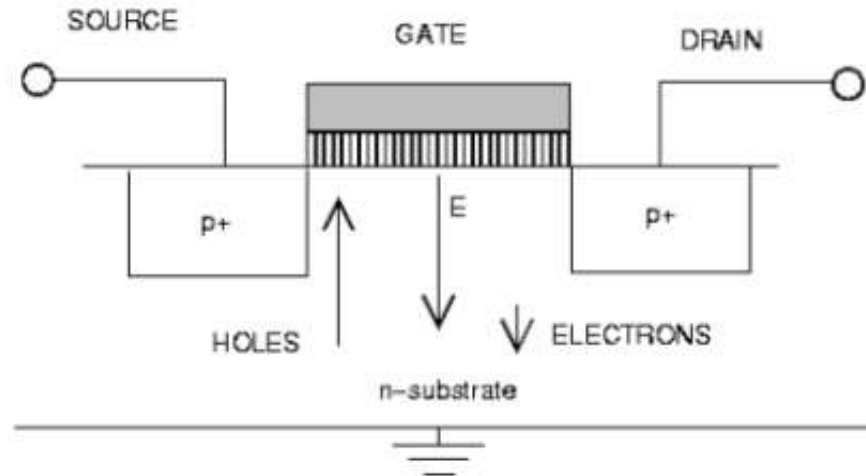
nMOS Saturation

- ❑ Channel pinches off
- ❑ I_{ds} independent of V_{ds}
- ❑ We say current *saturates*
- ❑ Similar to current source

$$\begin{aligned}V_{ds} - V_{gs} &> -V_t \\V_t &> V_{gs} - V_{ds} \\V_t &> V_{gd} \rightarrow \underline{V_{gd} < V_t}\end{aligned}$$



The pMOS transistor

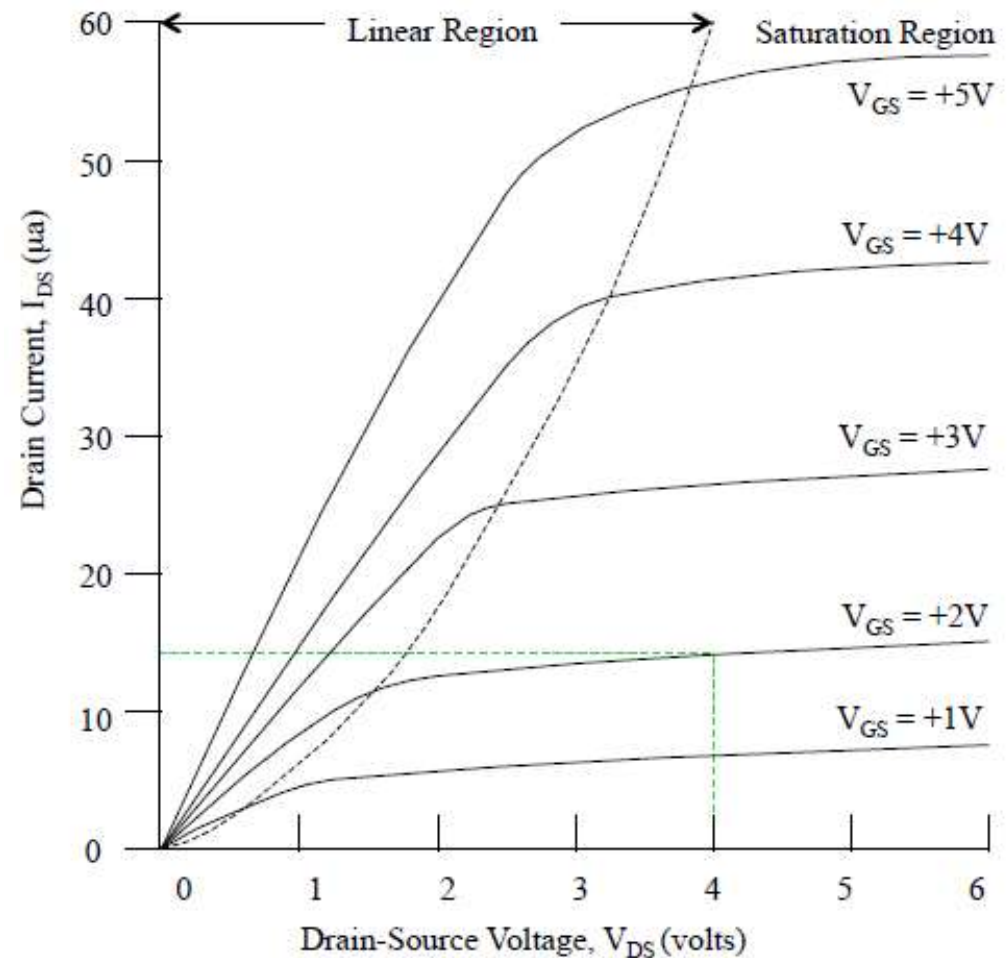


Moderately doped n- type substrate (or well) in which two heavily doped p+ regions, the **Source** and **Drain** are diffused

- Application of a negative gate voltage (w.r.t. source) draws holes into the region below the gate; channel changes from n to p-type (source-drain conduction path)
- Conduction due to holes; negative V_d sweeps holes from source (through channel) to drain

2. I-V Characteristics

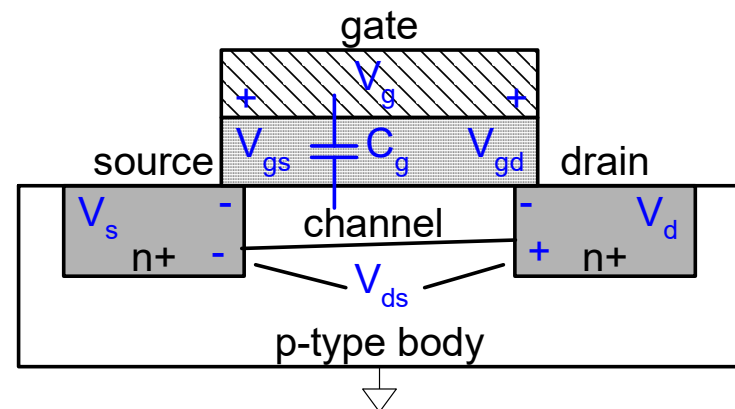
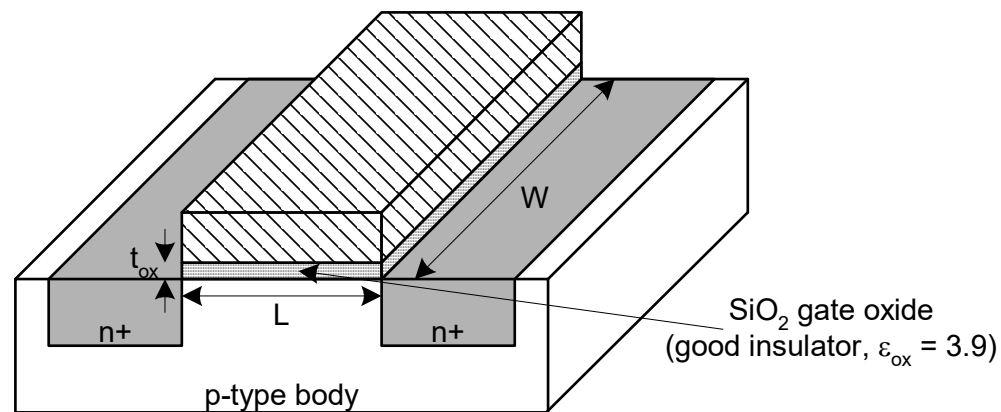
- In Linear region, I_{ds} depends on
 - How much charge is in the channel?
 - How fast is the charge moving?



Channel Charge

- ❑ MOS structure looks like parallel plate capacitor while operating in inversions
 - Gate – oxide – channel

- ❑ $Q_{\text{channel}} =$



Channel Charge

- MOS structure looks like parallel plate capacitor while operating in inversion:

- Gate – oxide – channel

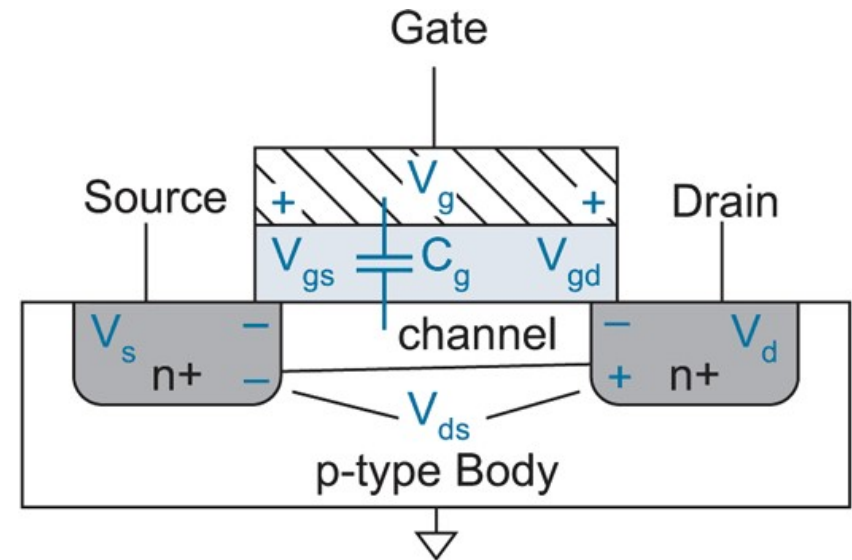
- $Q_{\text{channel}} = CV$

- $C = C_g = \epsilon_{\text{ox}} WL / t_{\text{ox}} = c_{\text{ox}} WL$

- $V = V_{gc} - V_t = (V_{gs} - V_{ds}/2) - V_t$

$$c_{\text{ox}} = \epsilon_{\text{ox}} / t_{\text{ox}}$$

$$V_{gc} = \frac{V_{gs} + V_{gd}}{2} = \frac{V_{gs} + V_{gs} - V_{ds}}{2} = V_{gs} - \frac{V_{ds}}{2}$$



Average gate to channel potential:

$$V_{gc} = (V_{gs} + V_{gd})/2 = V_{gs} - V_{ds}/2$$

FIG 2.5 Average gate to channel voltage

Carrier velocity

- ❑ Charge is carried by e-
- ❑ Electrons are propelled by the lateral electric field between source and drain
 - $E =$
- ❑ Carrier velocity v proportional to lateral E-field
 - $v =$
- ❑ Time for carrier to cross channel:
 - $t =$

nMOS Linear I-V

Now we know:

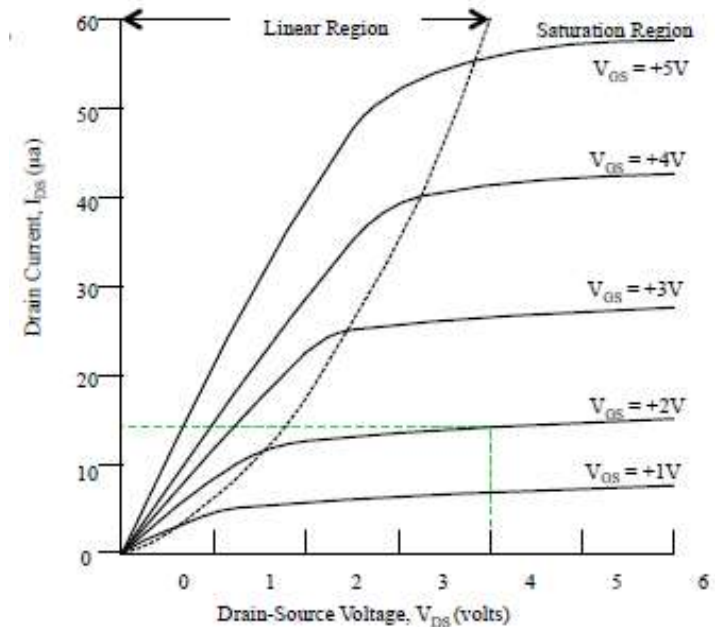
How much charge Q_{channel} is in the channel

How much time t each carrier takes to cross

$$I_{ds} =$$

=

=

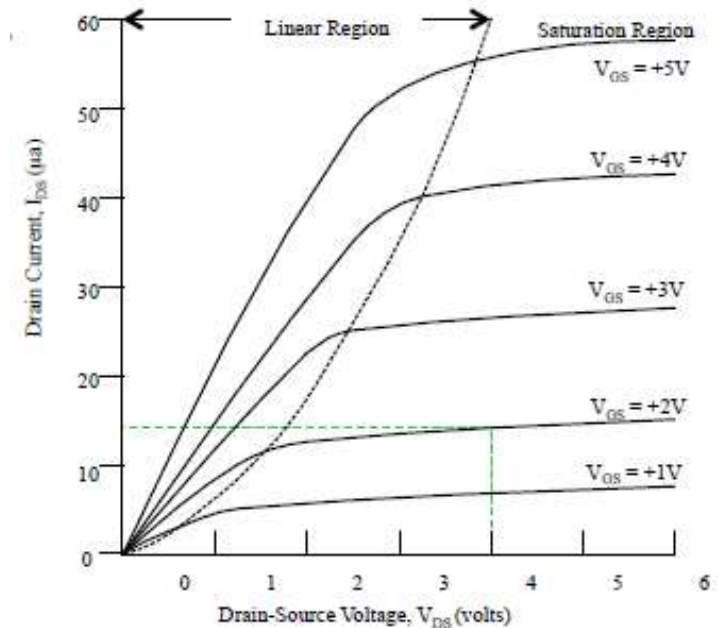


nMOS Saturation I-V

- If $V_{gd} < V_t$, channel pinches off near drain
 - When $V_{ds} > V_{dsat} =$
- Now drain voltage no longer increases current

$$I_{ds} =$$

$$=$$

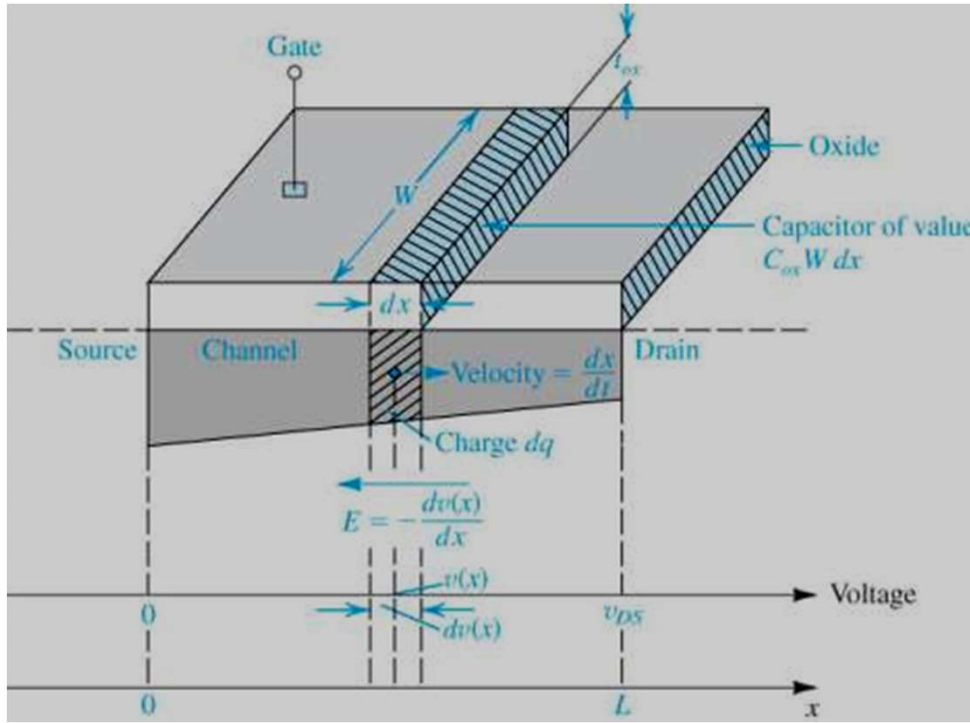


nMOS I-V Summary

- Shockley 1st order transistor models

$$I_{ds} = \begin{cases} 0 & V_{gs} < V_t & \text{cutoff} \\ \beta \left(V_{gs} - V_t - \frac{V_{ds}}{2} \right) V_{ds} & V_{ds} < V_{dsat} & \text{linear} \\ \frac{\beta}{2} (V_{gs} - V_t)^2 & V_{ds} > V_{dsat} & \text{saturation} \end{cases}$$

Tóm tắt cách tìm quan hệ dòng và áp trong N-EMOS



$$i = \frac{dq}{dt} = \frac{dq}{dx} \frac{dx}{dt} \quad (1)$$

$$Q = CV \quad C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} \quad (2)$$

$$\epsilon_{ox} = 3.9\epsilon_0 = 3.9 \times 8.854 \times 10^{-12} = 3.45 \times 10^{-11} \text{ F/m}$$

$$dq = -C_{ox}(Wdx)[v_{GS} - v(x) - V_t] \quad (3)$$

$$E(x) = -\frac{dv(x)}{dx} \quad (\because E = -\nabla V)$$

$$\frac{dx}{dt} = -\mu_n E(x) = \mu_n \frac{dv(x)}{dx} \quad (4)$$

$$i = -\mu_n C_{ox} W [v_{GS} - v(x) - V_t] \frac{dv(x)}{dx}$$

$$i_D = -i = \mu_n C_{ox} W [v_{GS} - v(x) - V_t] \frac{dv(x)}{dx}$$

$$i_D dx = \mu_n C_{ox} W [v_{GS} - V_t - v(x)] dv(x)$$

$$\int_0^L i_D dx = \int_0^{v_{DS}} \mu_n C_{ox} W [v_{GS} - V_t - v(x)] dv(x)$$

$$i_D = (\mu_n C_{ox}) \left(\frac{W}{L} \right) \left[(v_{GS} - V_t) v_{DS} - \frac{1}{2} v_{DS}^2 \right] \quad (5)$$

Khi vào miền bão hòa, $v_{DS} = v_{GS} - V_t$

$$i_D = \frac{1}{2} (\mu_n C_{ox}) \left(\frac{W}{L} \right) (v_{GS} - V_t)^2 \quad (6)$$

$$k'_n = \mu_n C_{ox} \quad (7)$$

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} \quad (2)$$

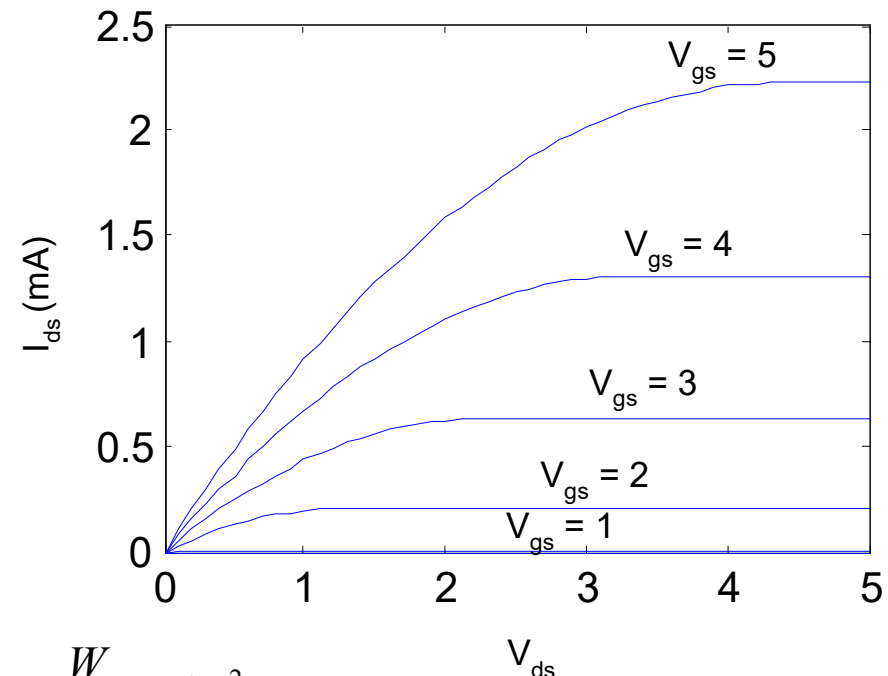
$$i_D = k'_n \left(\frac{W}{L} \right) \left[(v_{GS} - V_t) v_{DS} - \frac{1}{2} v_{DS}^2 \right] \quad (\text{Triode region}) \quad (5a)$$

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (v_{GS} - V_t)^2 \quad (\text{saturation region}) \quad (6a)$$

$\left(\frac{W}{L} \right)$: Aspect ratio of the MOSFET
(Tỉ số hình dạng của MOSFET)

Example

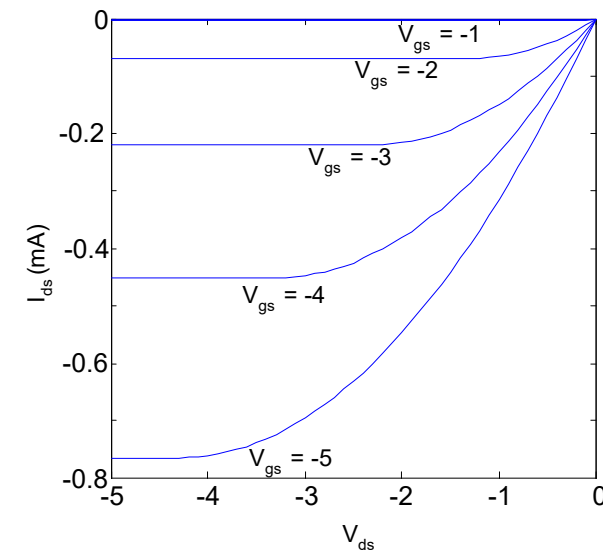
- ❑ We will be using a 0.6 μm process for your project
 - From AMI Semiconductor
 - $t_{\text{ox}} = 100 \text{ \AA}$
 - $\mu = 350 \text{ cm}^2/\text{V}\cdot\text{s}$
 - $V_t = 0.7 \text{ V}$
- ❑ Plot I_{ds} vs. V_{ds}
 - $V_{\text{gs}} = 0, 1, 2, 3, 4, 5$
 - Use $W/L = 4/2 \lambda$



$$\beta = \mu C_{\text{ox}} \frac{W}{L} = (350) \left(\frac{3.9 \times 8.85 \cdot 10^{-14}}{100 \cdot 10^{-8}} \right) \left(\frac{W}{L} \right) = 120 \frac{W}{L} \mu\text{A}/\text{V}^2$$

pMOS I-V

- ❑ All dopings and voltages are inverted for pMOS
 - Source is the more positive terminal
- ❑ Mobility μ_p is determined by holes
 - Typically 2-3x lower than that of electrons μ_n
 - 120 cm²/V•s in AMI 0.6 μ m process
- ❑ Thus pMOS must be wider to provide same current
 - In this class, assume
$$\mu_n / \mu_p = 2$$



Assignment

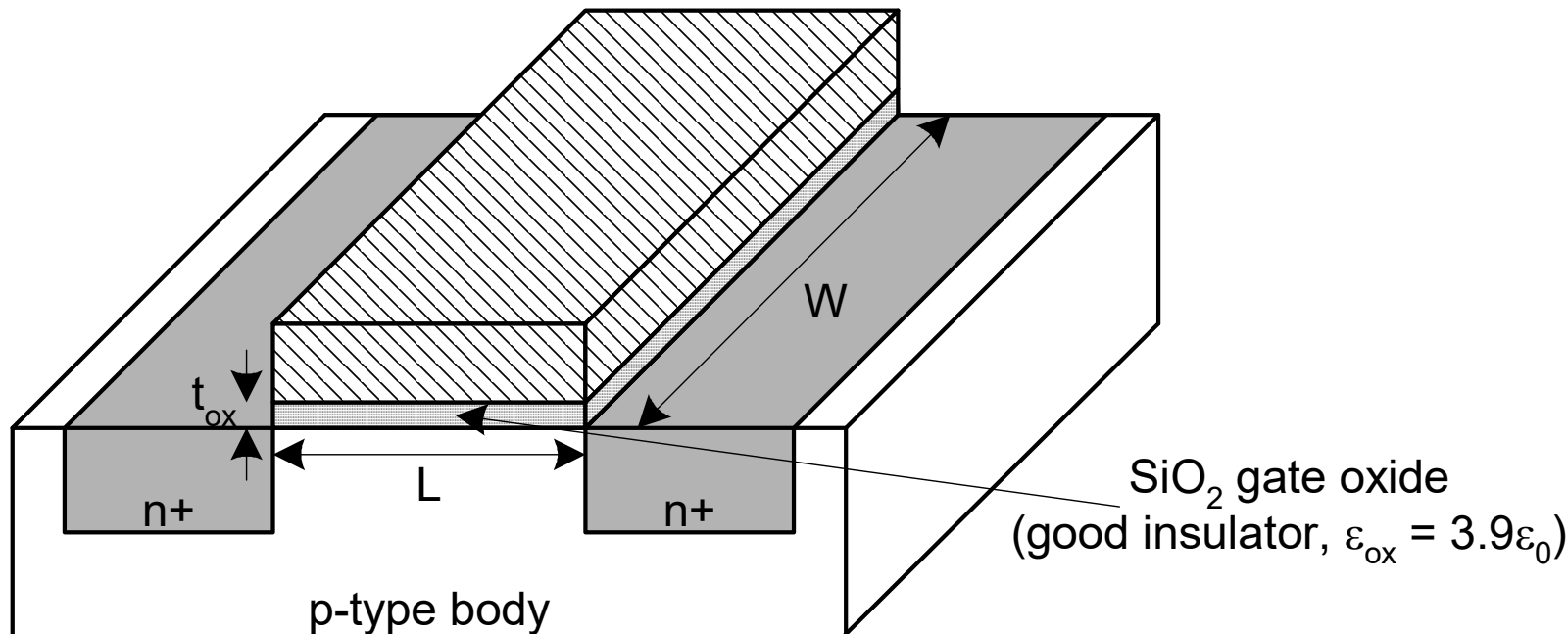
- Consider an nMOS transistor in a $0.6\text{ }\mu\text{m}$ process with $W/L = 4/2\text{ }\lambda$ (i.e., $1.2/0.6\text{ }\mu\text{m}$). In this process, the gate oxide thickness is $100\text{ }\text{\AA}$ and the mobility of electrons is $350\text{ cm}^2/\text{V}\cdot\text{s}$. The threshold voltage is 0.7 V . Plot I_{ds} vs. V_{ds} for $V_{gs} = 0, 1, 2, 3, 4,$ and 5 V .

3. C-V Characteristics

- ❑ Any two conductors separated by an insulator have capacitance
- ❑ Gate to channel capacitor is very important
 - Creates channel charge necessary for operation
- ❑ Source and drain have capacitance to body
 - Across reverse-biased diodes
 - Called diffusion capacitance because it is associated with source/drain diffusion

Gate Capacitance

- ❑ Approximate channel as connected to source
- ❑ $C_g = C_{ox}WL = C_{permicron}W$
- ❑ $C_{permicron} = C_{ox}L = \epsilon_{ox}L/t_{ox}$
- ❑ $C_{permicron}$ is typically about 2 fF/ μm



Gate Capacitance

- When the transistor is off, the channel is not inverted
 $C_g = C_{gb} = \epsilon_{ox} WL / t_{ox} = C_{ox} WL$
- Let's call $C_{ox} WL = C_0$
- When the transistor is on, the channel extends from the source to the drain (if the transistor is unsaturated, or to the pinchoff point otherwise)
 $C_g = C_{gb} + C_{gs} + C_{gd}$

Gate Capacitance

Table 2.1 Approximation of intrinsic MOS gate capacitance

Parameter	Cutoff	Linear	Saturation
C_{gb}	C_0	0	0
C_{gs}	0	$C_0/2$	$2/3 C_0$
C_{gd}	0	$C_0/2$	0
$C_g = C_{gs} + C_{gd} + C_{gb}$	C_0	C_0	$2/3 C_0$

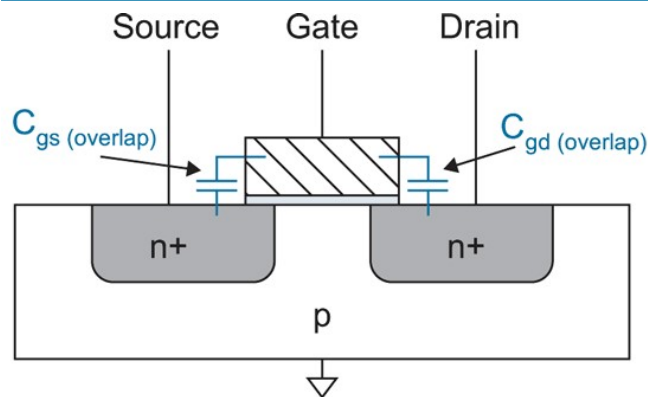


FIG 2.10 Overlap capacitance

In reality the gate overlaps source and drain. Thus, the gate capacitance should include not only the intrinsic capacitance but also parasitic overlap capacitances:

$$C_{gs}(\text{overlap}) = C_{ox} W L_D$$

$$C_{gd}(\text{overlap}) = C_{ox} W L_D$$

Device Capacitances, Cont'd

Off Region

- $V_{gs} \leq V_t$; when the MOS device is off, only C_{gb} (due to the series combination of gate oxide and depletion layer capacitance) is non-zero.
- $C_{gb} = C_{ox} = \epsilon A / t_{ox}$, where A is the gate area, and $\epsilon = \epsilon_0 \epsilon_{SiO_2}$
 - ϵ_0 is the permittivity of free space (8.854×10^{-12} F/m), and ϵ_{SiO_2} is the dielectric constant of SiO_2 (about 3.9)

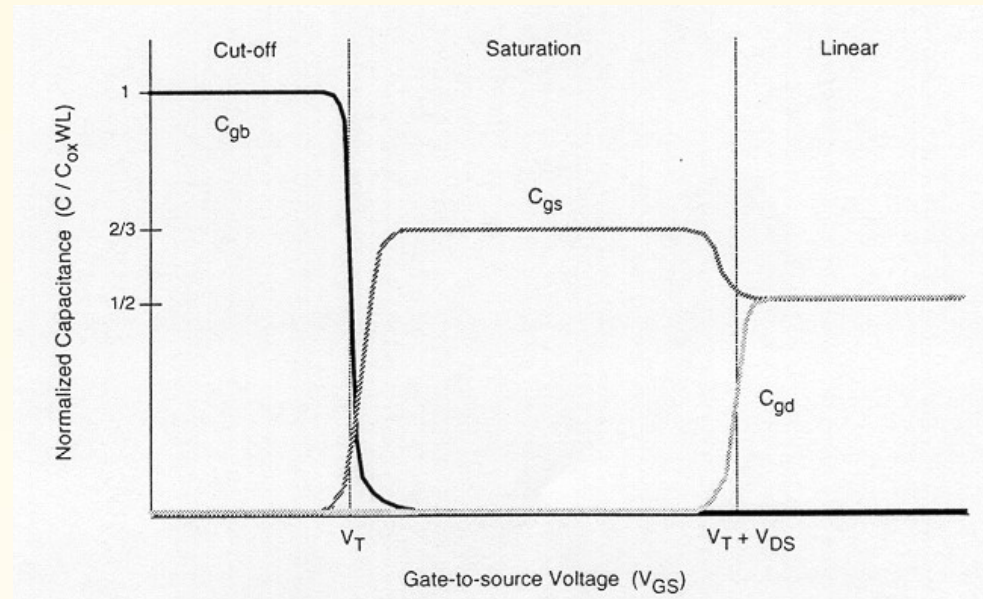
Linear Region

- Depletion region exists, forming dielectric of depletion capacitance, C_{dep} in series with C_{ox}
- As the device turns on, C_{gb} reduces to 0
- The gate capacitance is now a function of the gate voltage

Device Capacitances, Cont'd

Saturated Region

- Region under the gate is heavily inverted, and drain region of channel pinched off, with C_{gd} reducing to zero
- Gate capacitance is now less than C_{ox}



Approximation of Gate Capacitance

- For simplicity, we can assume the gate capacitance to be constant, $C_g = A/t_{ox}$

Diffusion (Source/Drain) Capacitance

Capacitance at the drain (C_{db}) or source (C_{sb}) of a device, or when diffusion is used as a wire

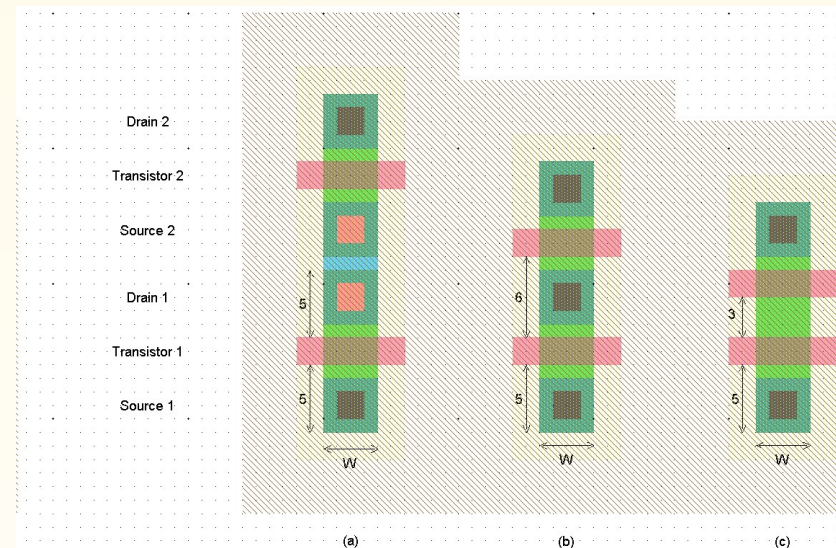
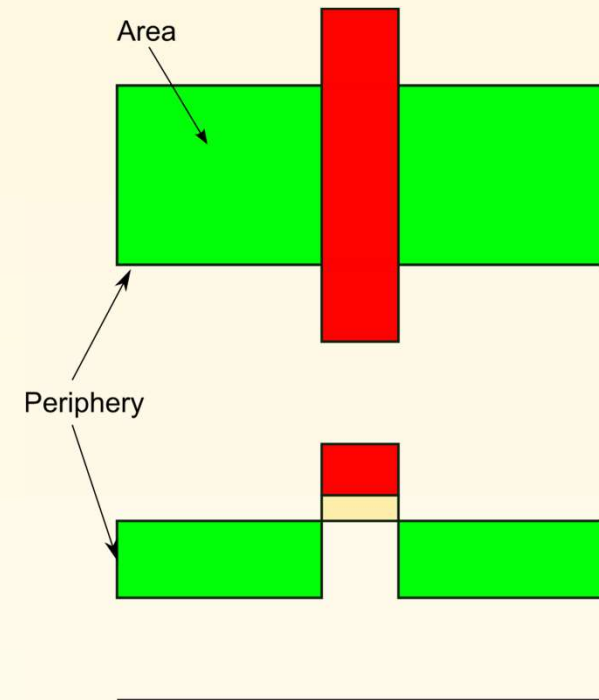
Two components:

- 1 An **Area** component
- 2 A **Peripheral** (sidewall) component

The peripheral component comes from the depth of the diffusion

Assume diffusion capacitance is **approximately C_g for contacted diffusion**

It is **$1/2 C_g$ for uncontacted diffusion**



Detailed Gate Capacitance

Capacitance	Cutoff	Linear	Saturation
C_{gb} (total)	C_0	0	0
C_{gd} (total)	$C_{ox}WL_D$	$C_0/2 + C_{ox}WL_D$	$C_{ox}WL_D$
C_{gs} (total)	$C_{ox}WL_D$	$C_0/2 + C_{ox}WL_D$	$2/3 C_0 + C_{ox}WL_D$

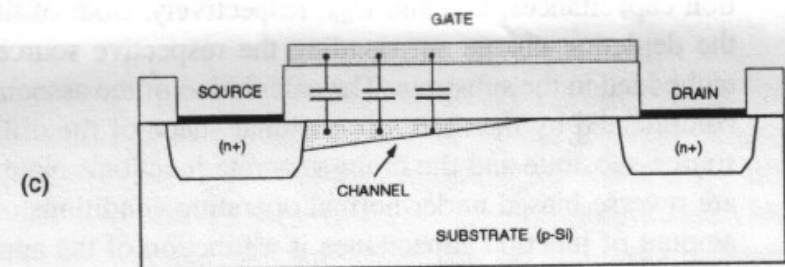
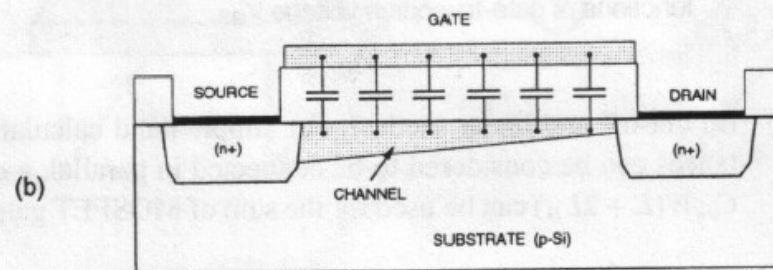
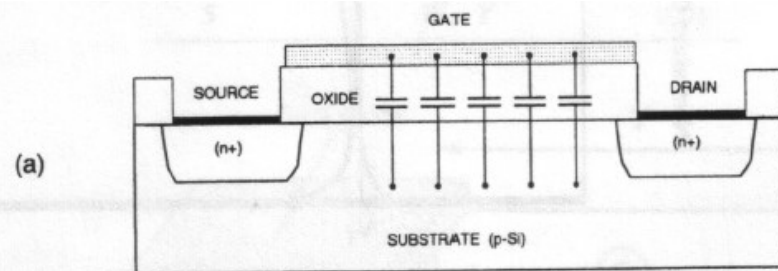
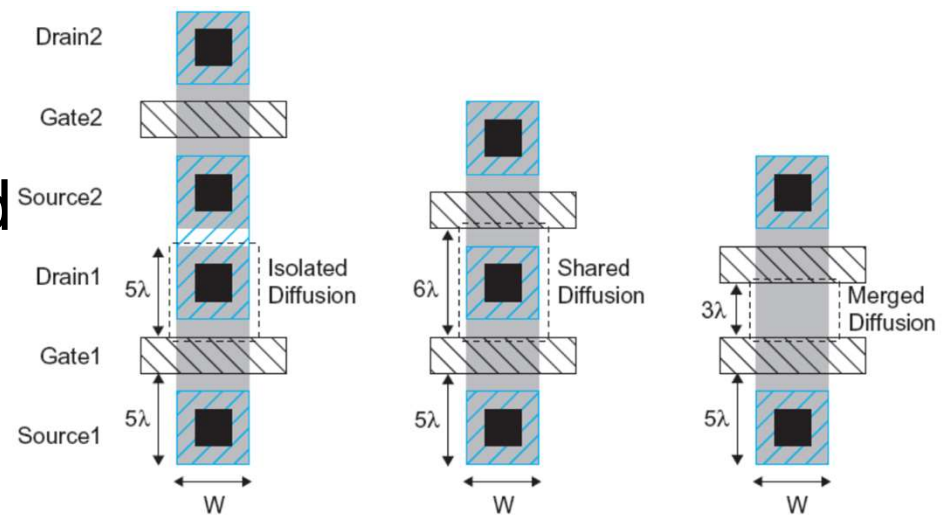


Figure 3.31 Schematic representation of MOSFET oxide capacitances during (a) cut-off, (b) linear, and (c) saturation modes.

Diffusion Capacitance

- ❑ C_{sb} , C_{db}
- ❑ Undesirable, called *parasitic* capacitance
- ❑ Capacitance depends on area and perimeter
 - Use small diffusion nodes
 - Comparable to C_g for contacted diff
 - $\frac{1}{2} C_g$ for uncontacted
 - Varies with process



Lumped representation of the MOSFET capacitances

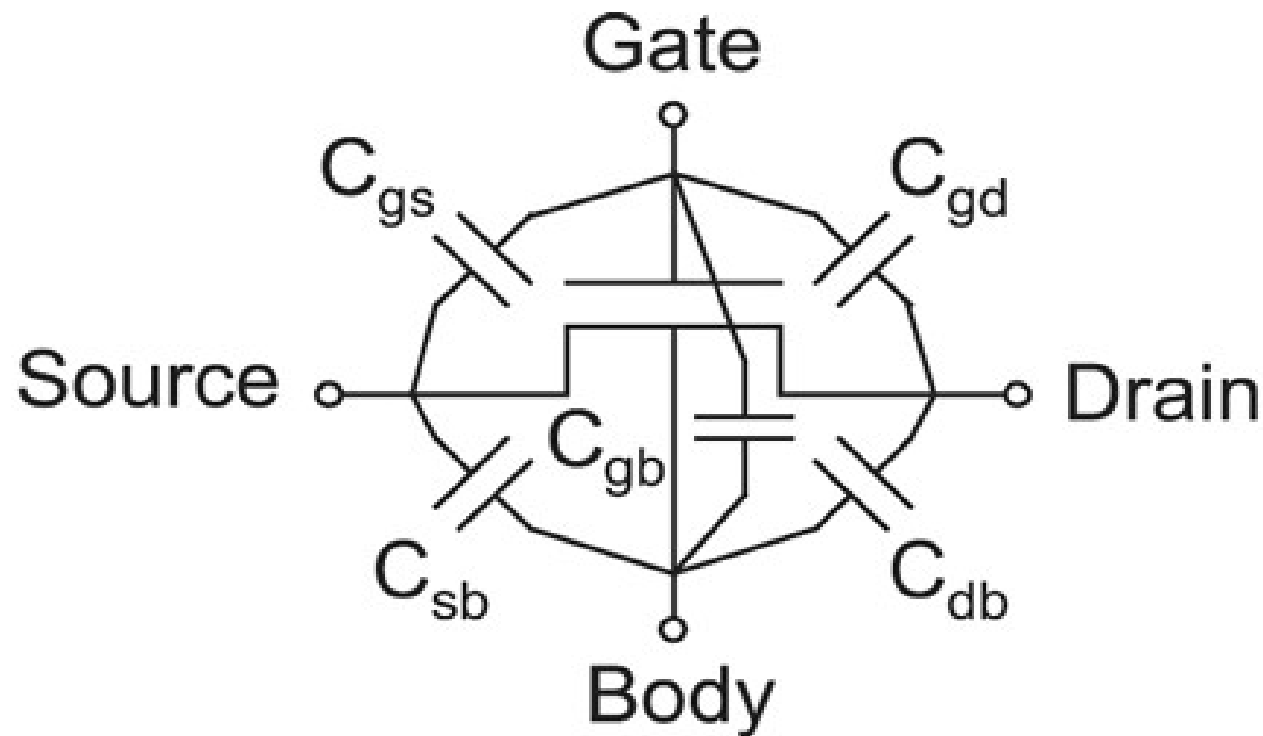


FIG 2.14 Capacitances of an MOS transistor

Review

1. Which factors affect to switching speed of MOS transistors?
2. What are three operation modes of MOS transistors?
3. When is a MOS transistor OFF?
4. When does a MOS transistor operate in saturation region?
5. What does the drain-source current I_{ds} depend on?
6. What does the gate capacitance C_g depend on?

4. DC Transfer Characteristics

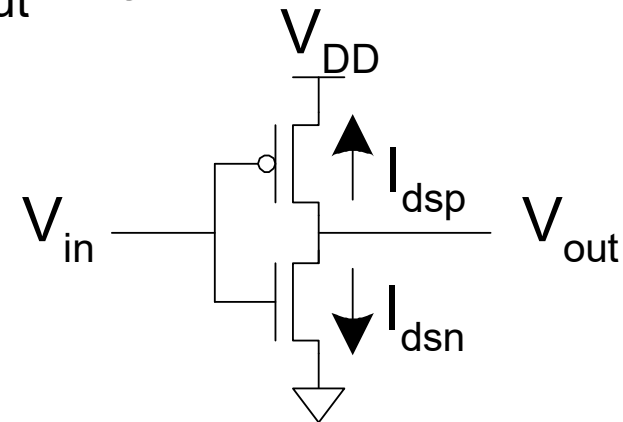
- ❑ Assuming the input changes **slowly enough** that capacitances have plenty of time to charge or discharge.
- ❑ Specific ranges of input and output voltages are defined as valid 0 and 1 logic levels

DC Response

□ DC Response: V_{out} vs. V_{in} for a gate

□ Ex: Inverter

- When $V_{in} = 0 \rightarrow V_{out} = V_{DD}$
- When $V_{in} = V_{DD} \rightarrow V_{out} = 0$
- In between, V_{out} depends on transistor size and current
- By KCL, must settle such that $I_{dsn} = |I_{dsp}|$
- We could solve equations
- But graphical solution gives more insight

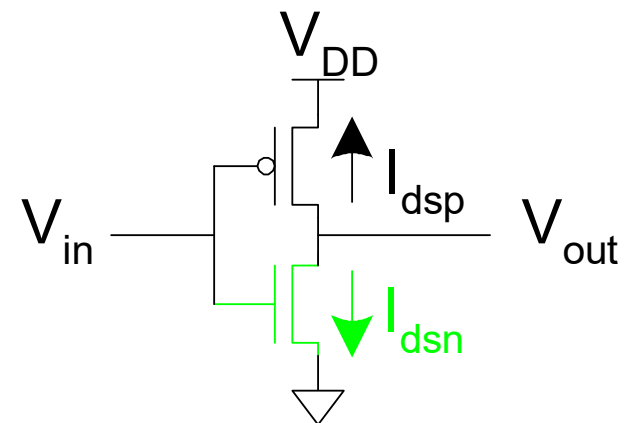


Transistor Operation

- ❑ Current depends on region of transistor behavior
- ❑ For what V_{in} and V_{out} are nMOS and pMOS in
 - Cutoff?
 - Linear?
 - Saturation?

nMOS Operation

Cutoff	Linear	Saturated
$V_{gsn} <$	$V_{gsn} >$ $V_{dsn} <$	$V_{gsn} >$ $V_{dsn} >$



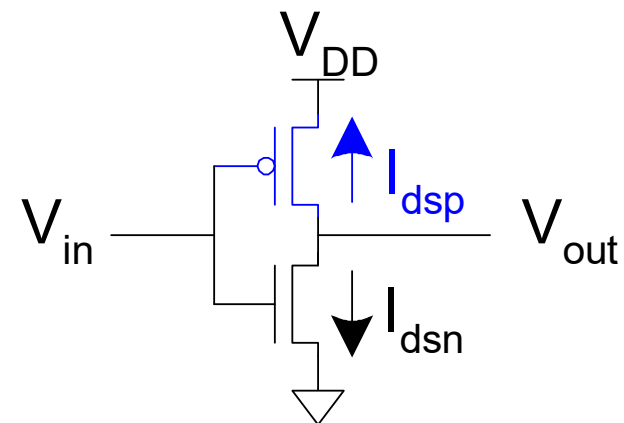
pMOS Operation

Cutoff	Linear	Saturated
$V_{gsp} > V_{tp}$ $V_{in} > V_{DD} + V_{tp}$	$V_{gsp} < V_{tp}$ $V_{in} < V_{DD} + V_{tp}$ $V_{dsp} > V_{gsp} - V_{tp}$ $V_{out} > V_{in} - V_{tp}$	$V_{gsp} < V_{tp}$ $V_{in} < V_{DD} + V_{tp}$ $V_{dsp} < V_{gsp} - V_{tp}$ $V_{out} < V_{in} - V_{tp}$

$$V_{gsp} = V_{in} - V_{DD}$$

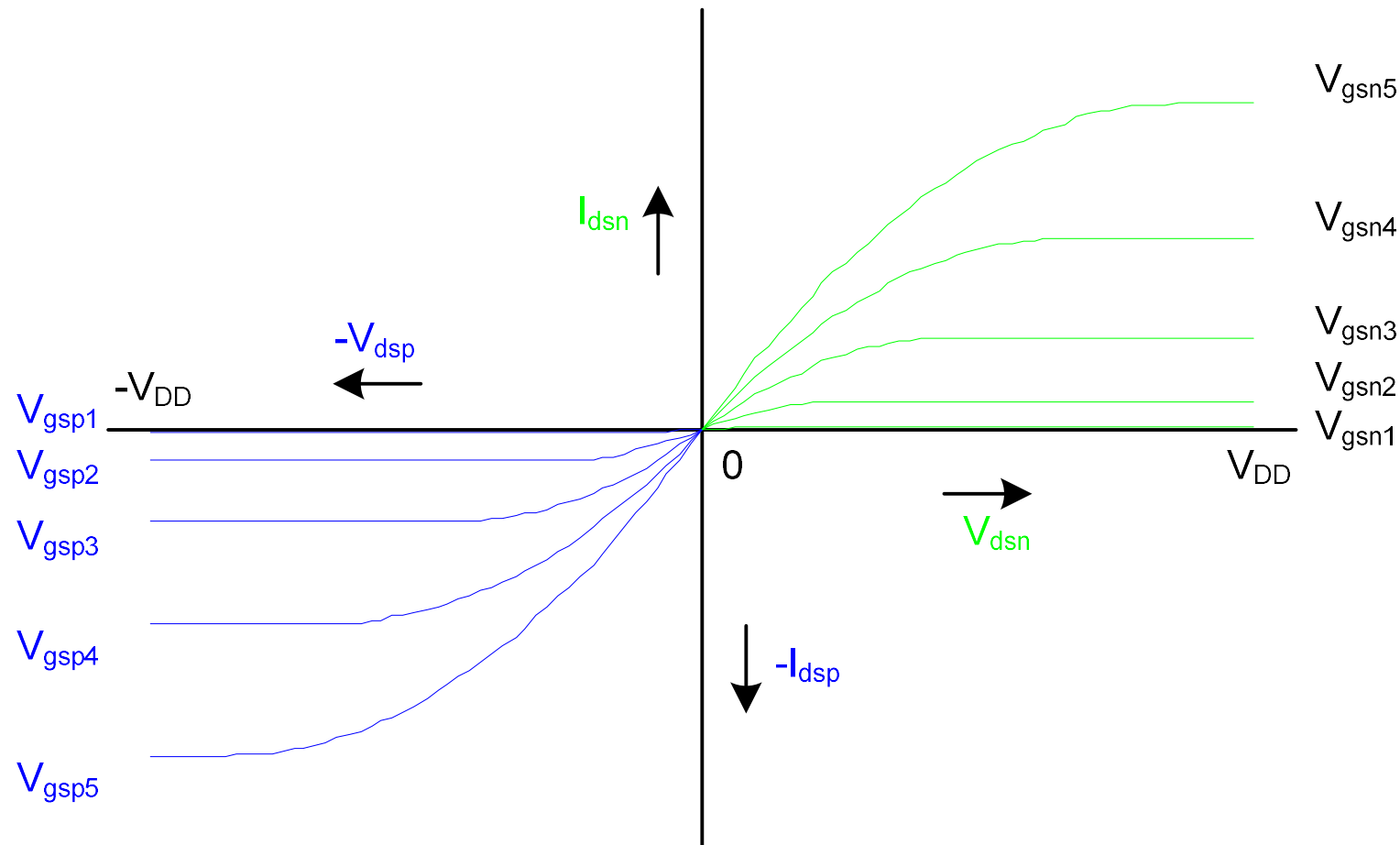
$$V_{tp} < 0$$

$$V_{dsp} = V_{out} - V_{DD}$$

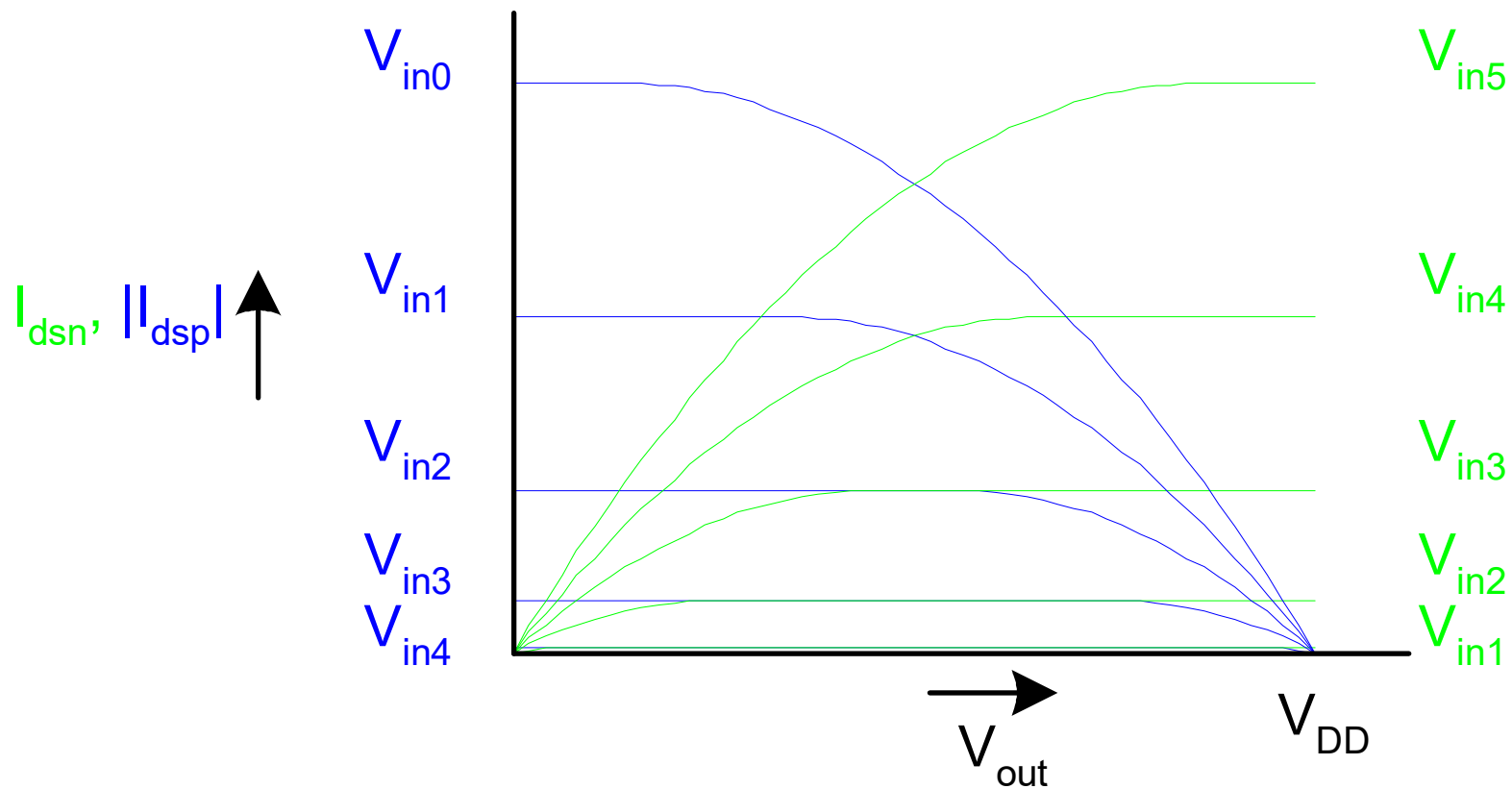


I-V Characteristics

- Make pMOS is wider than nMOS such that $\beta_n = \beta_p$

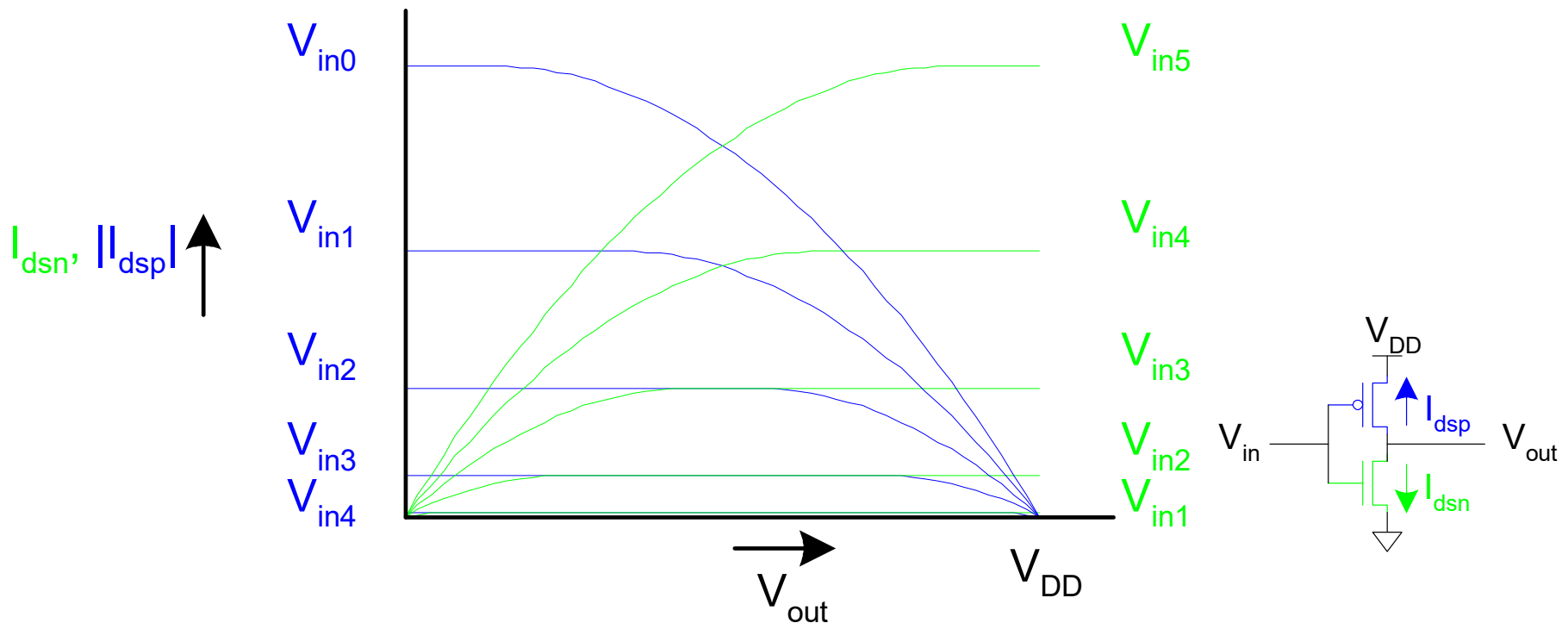


Current vs. V_{out} , V_{in}

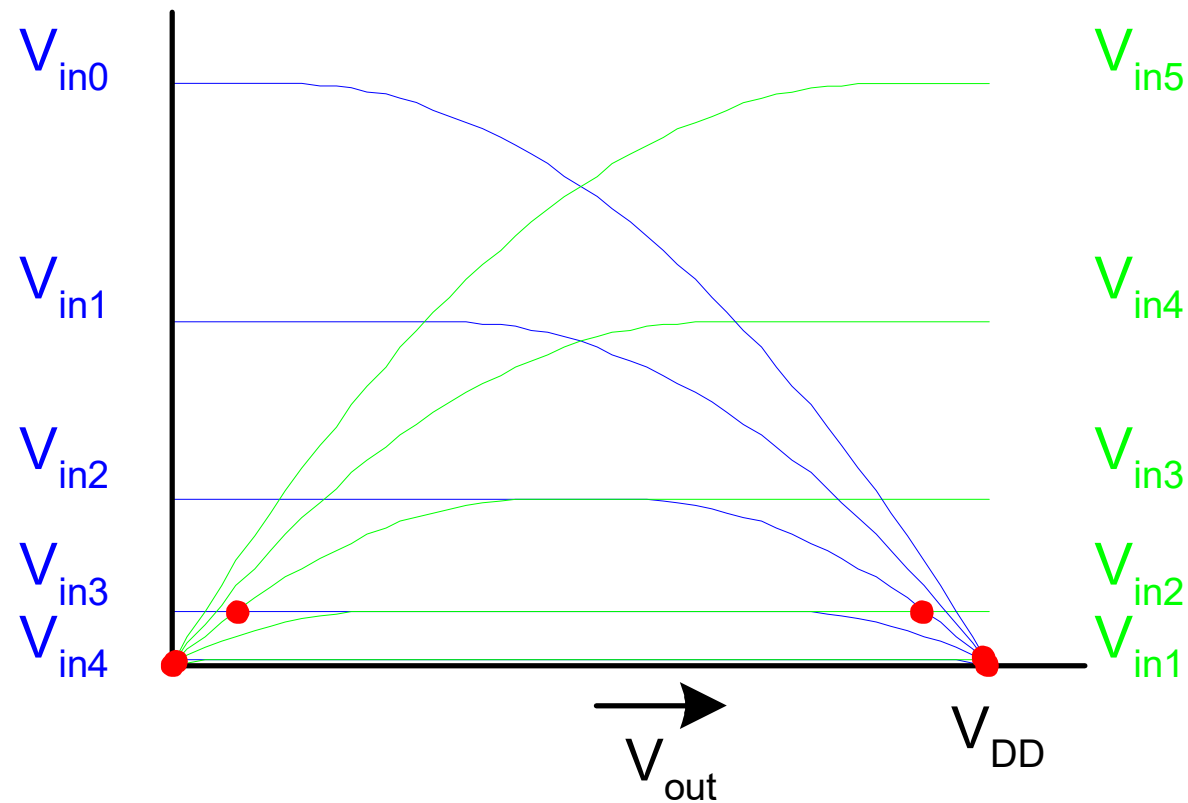


Load Line Analysis

- For a given V_{in} :
 - Plot I_{dsn} , I_{dsp} vs. V_{out}
 - V_{out} must be where |currents| are equal in

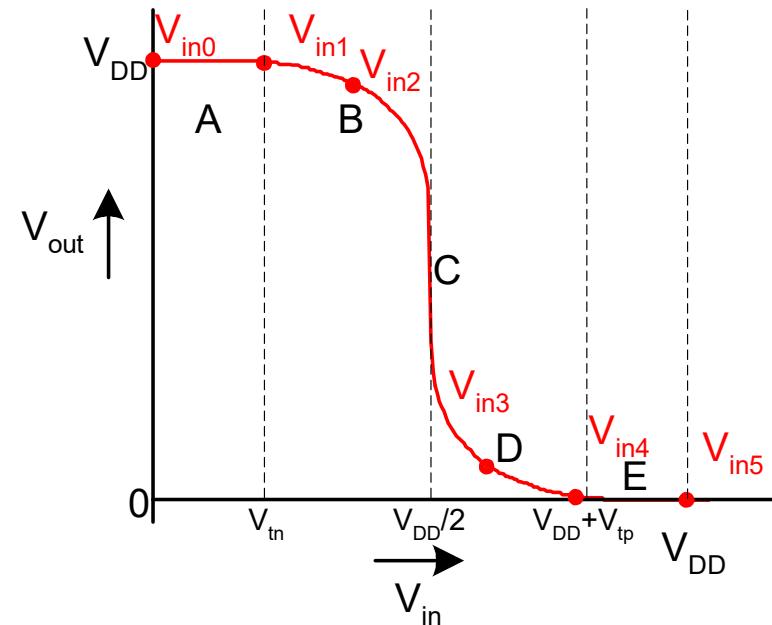
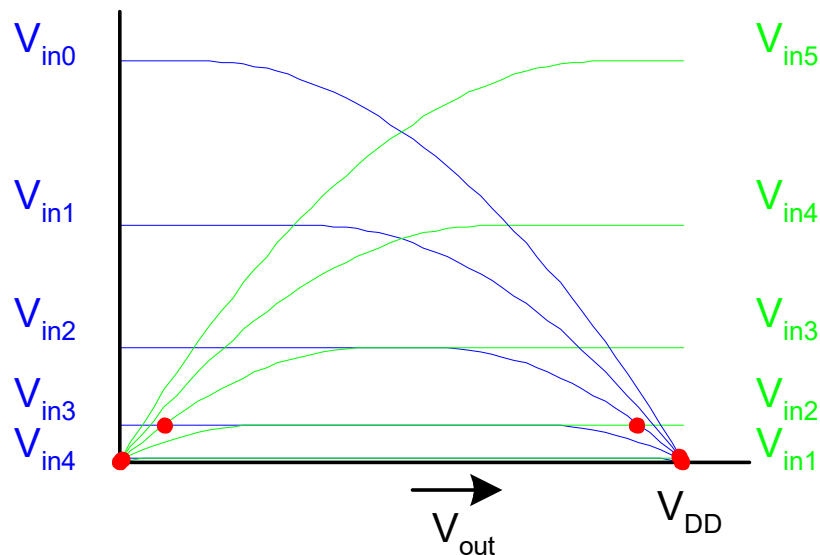


Load Line Analysis



DC Transfer Curve

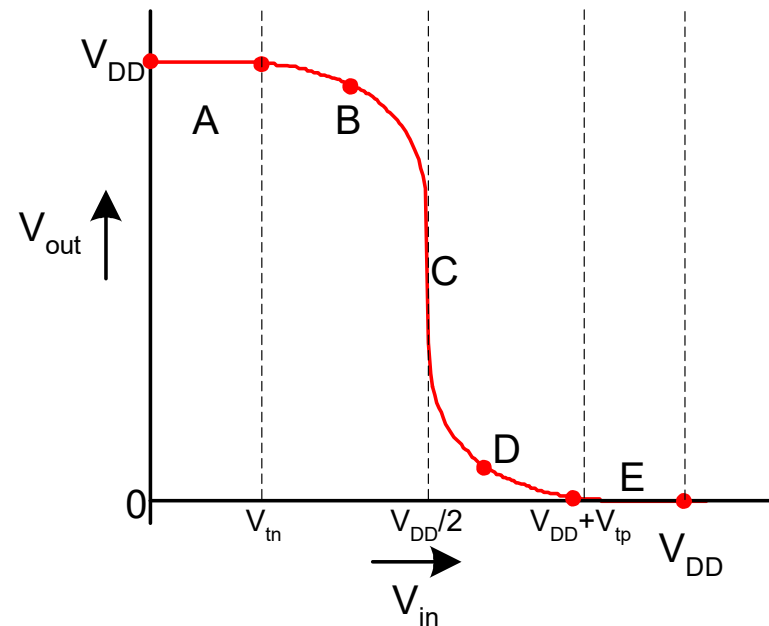
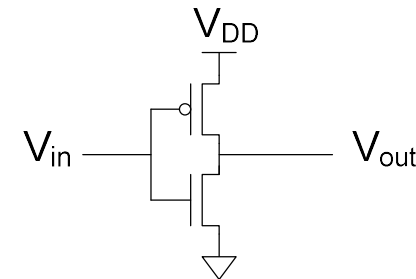
- Transcribe points onto V_{in} vs. V_{out} plot



Operating Regions

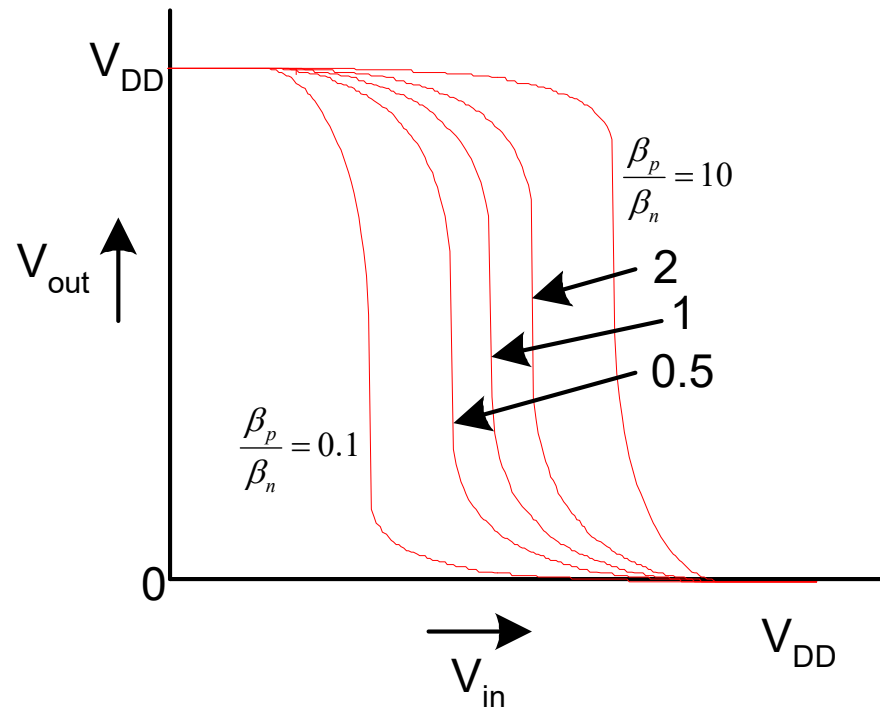
- ❑ Revisit transistor operating regions

Region	nMOS	pMOS
A		
B		
C		
D		
E		



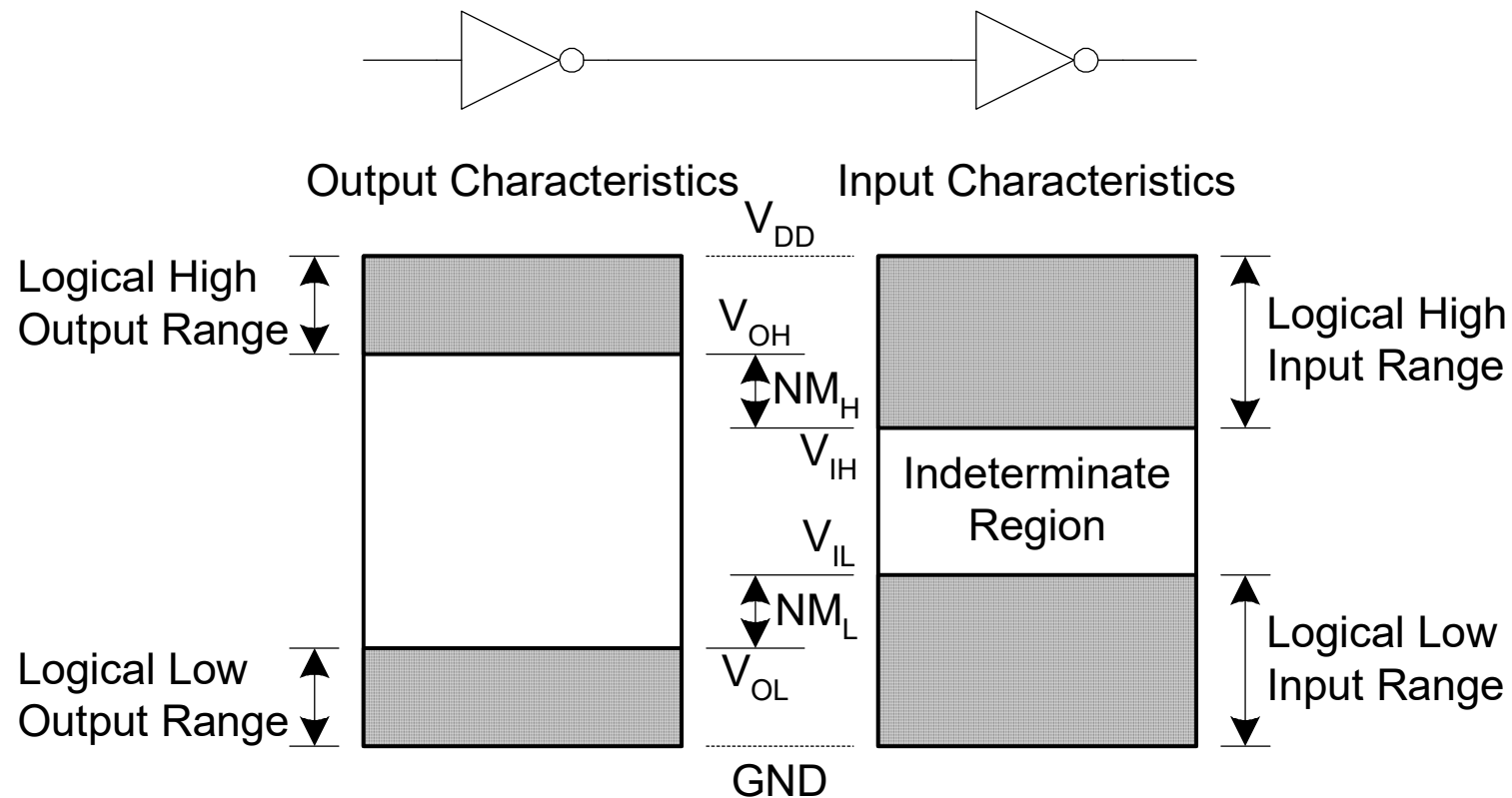
Beta Ratio

- ❑ If $\beta_p / \beta_n \neq 1$, switching point will move from $V_{DD}/2$
- ❑ Called *skewed gate*
- ❑ Other gates: collapse into equivalent inverter



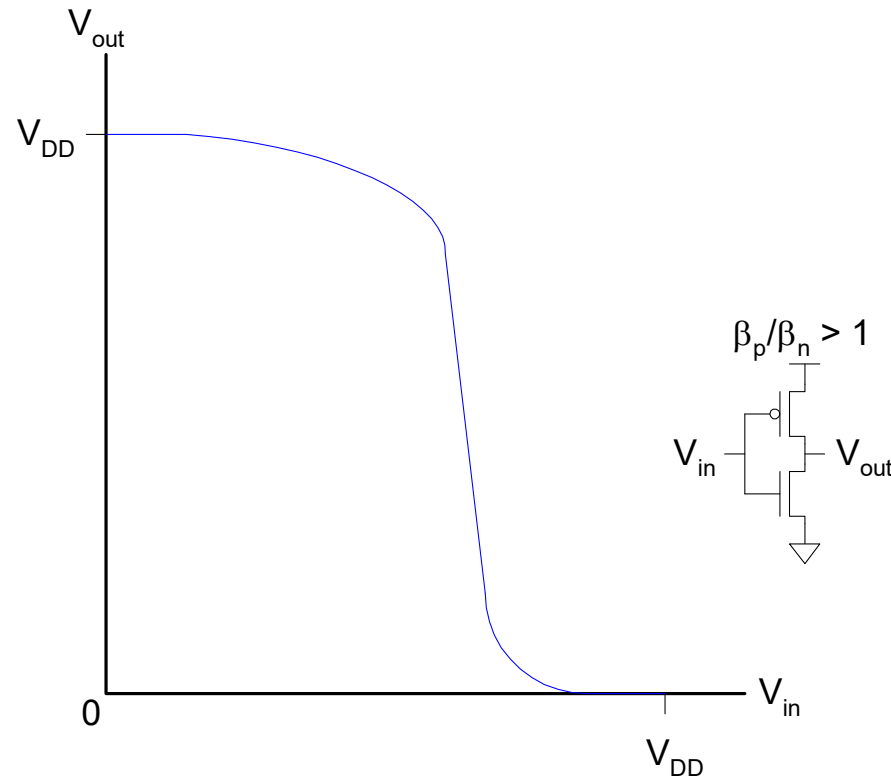
Noise Margins

- ❑ How much noise can a gate input see before it does not recognize the input?



Logic Levels

- ❑ To maximize noise margins, select logic levels at
 - unity gain point of DC transfer characteristic



Transient Response

- ❑ *DC analysis* tells us V_{out} if V_{in} is constant
- ❑ *Transient analysis* tells us $V_{out}(t)$ if $V_{in}(t)$ changes
 - Requires solving differential equations
- ❑ Input is usually considered to be a step or ramp
 - From 0 to V_{DD} or vice versa

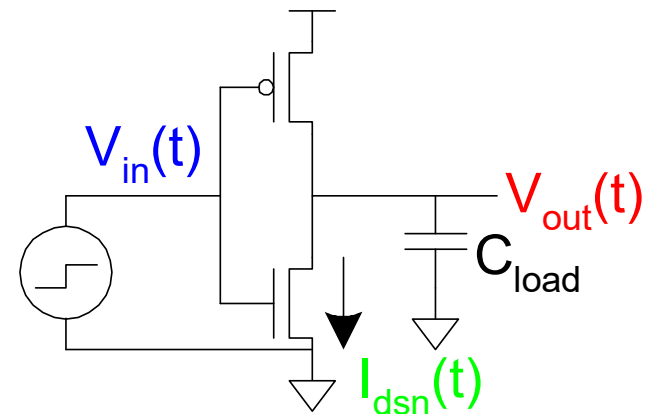
Inverter Step Response

□ Ex: find step response of inverter driving load cap

$$V_{in}(t) =$$

$$V_{out}(t < t_0) =$$

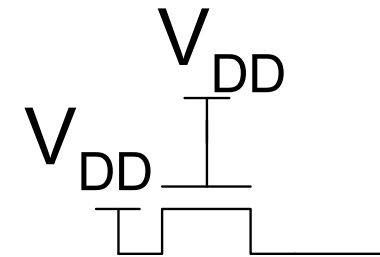
$$\frac{dV_{out}(t)}{dt} =$$



$$I_{dsn}(t) = \begin{cases} & t \leq t_0 \\ V_{out} > V_{DD} - V_t \\ V_{out} < V_{DD} - V_t \end{cases}$$

Pass Transistors

- ❑ We have assumed source is grounded
- ❑ What if source > 0 ?
 - e.g. pass transistor passing V_{DD}
- ❑ $V_g = V_{DD}$
 - If $V_s > V_{DD} - V_t$, $V_{gs} < V_t$
 - Hence transistor would turn itself off
- ❑ nMOS pass transistors pull no higher than $V_{DD} - V_{tn}$
 - Called a degraded “1”
 - Approach degraded value slowly (low I_{ds})
- ❑ pMOS pass transistors pull no lower than V_{tp}
- ❑ **Transmission gates** are needed to pass both 0 and 1



Pass Transistor Ckts

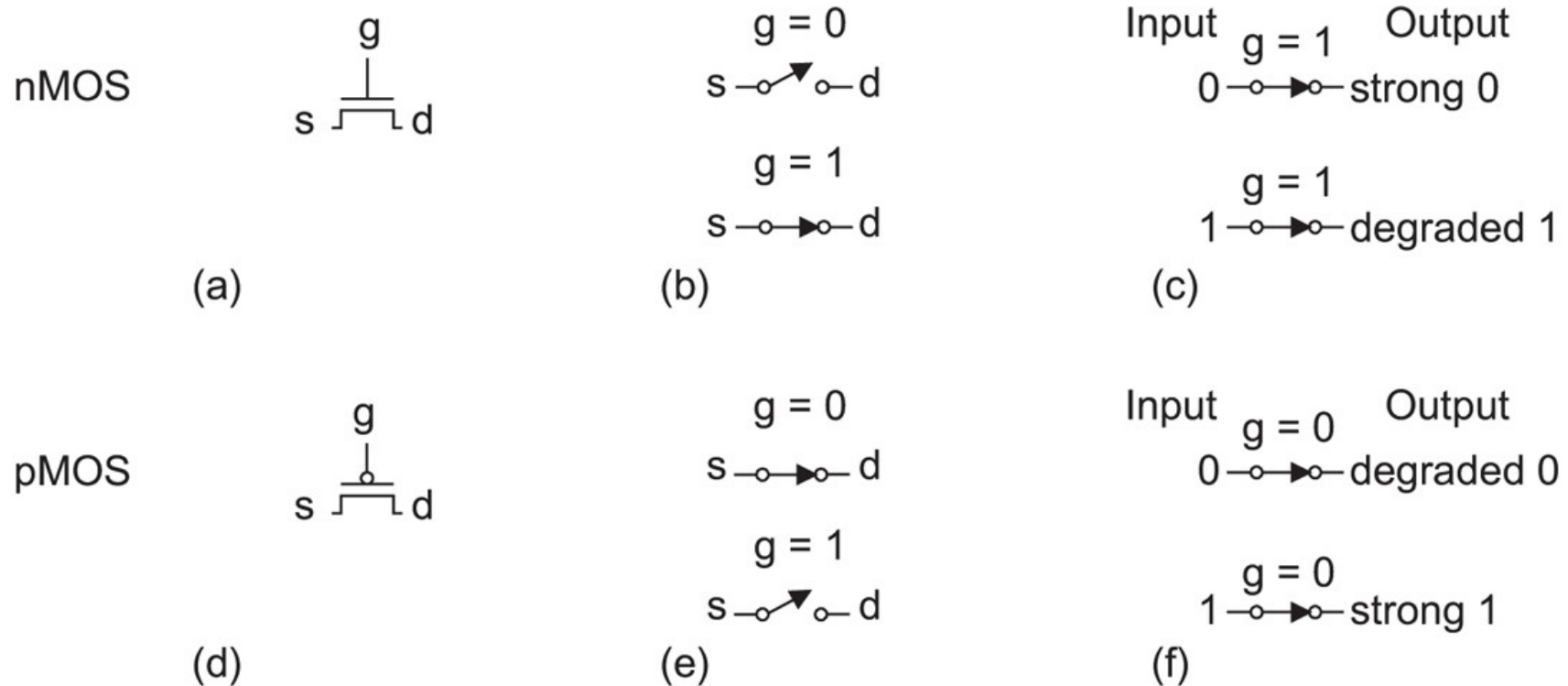
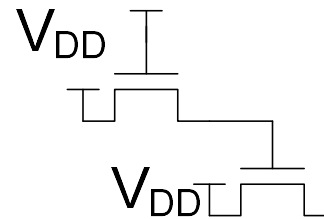
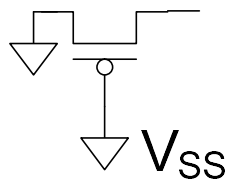
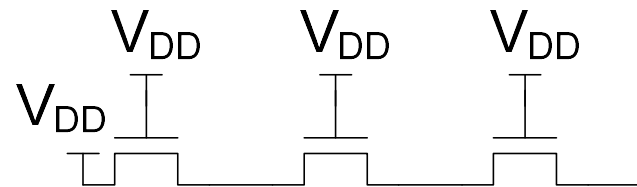
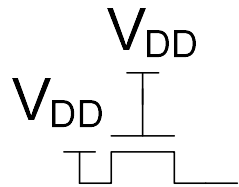
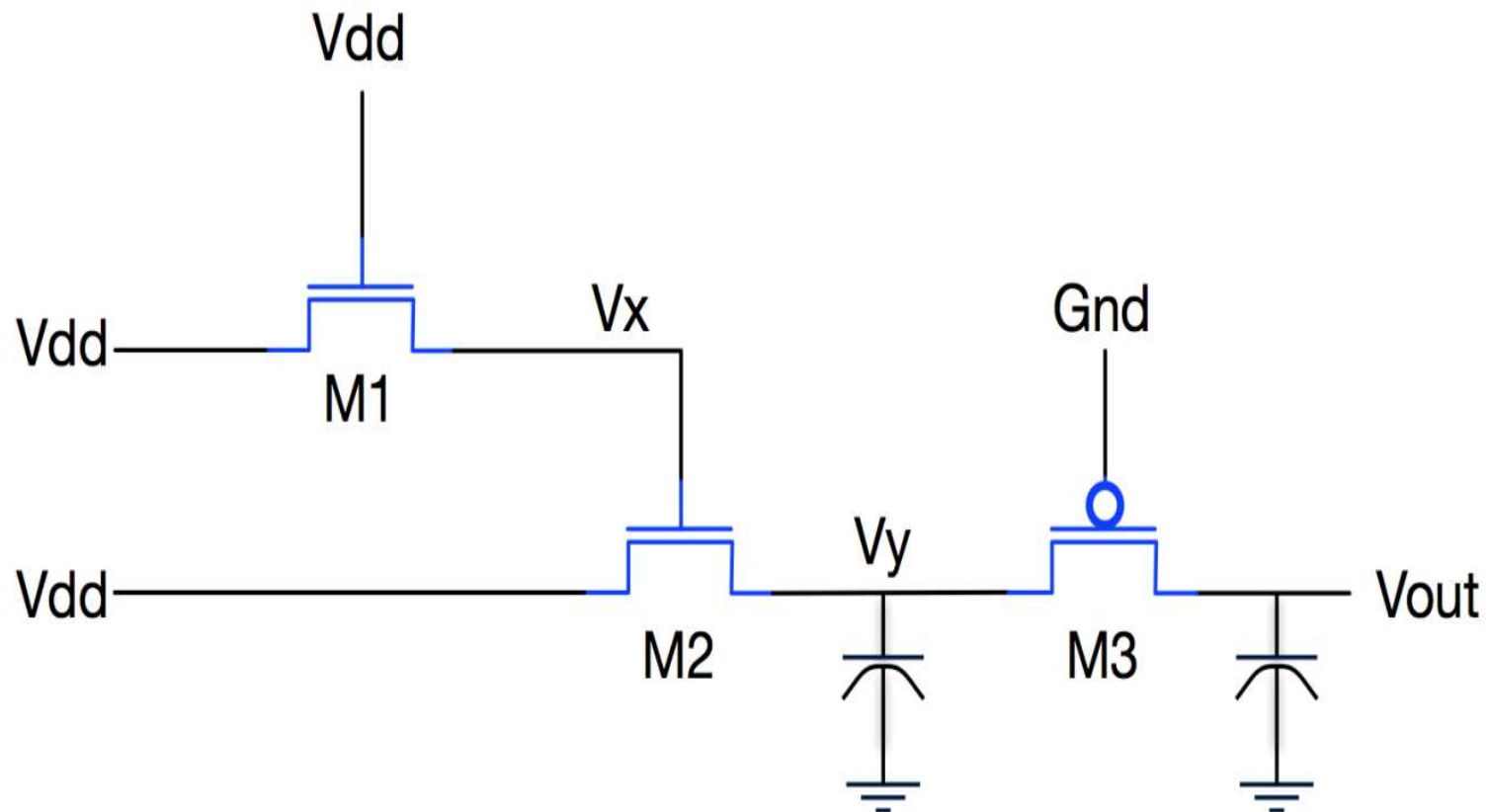


FIG 1.19 Pass transistor strong and degraded outputs

Pass Transistor Ckts



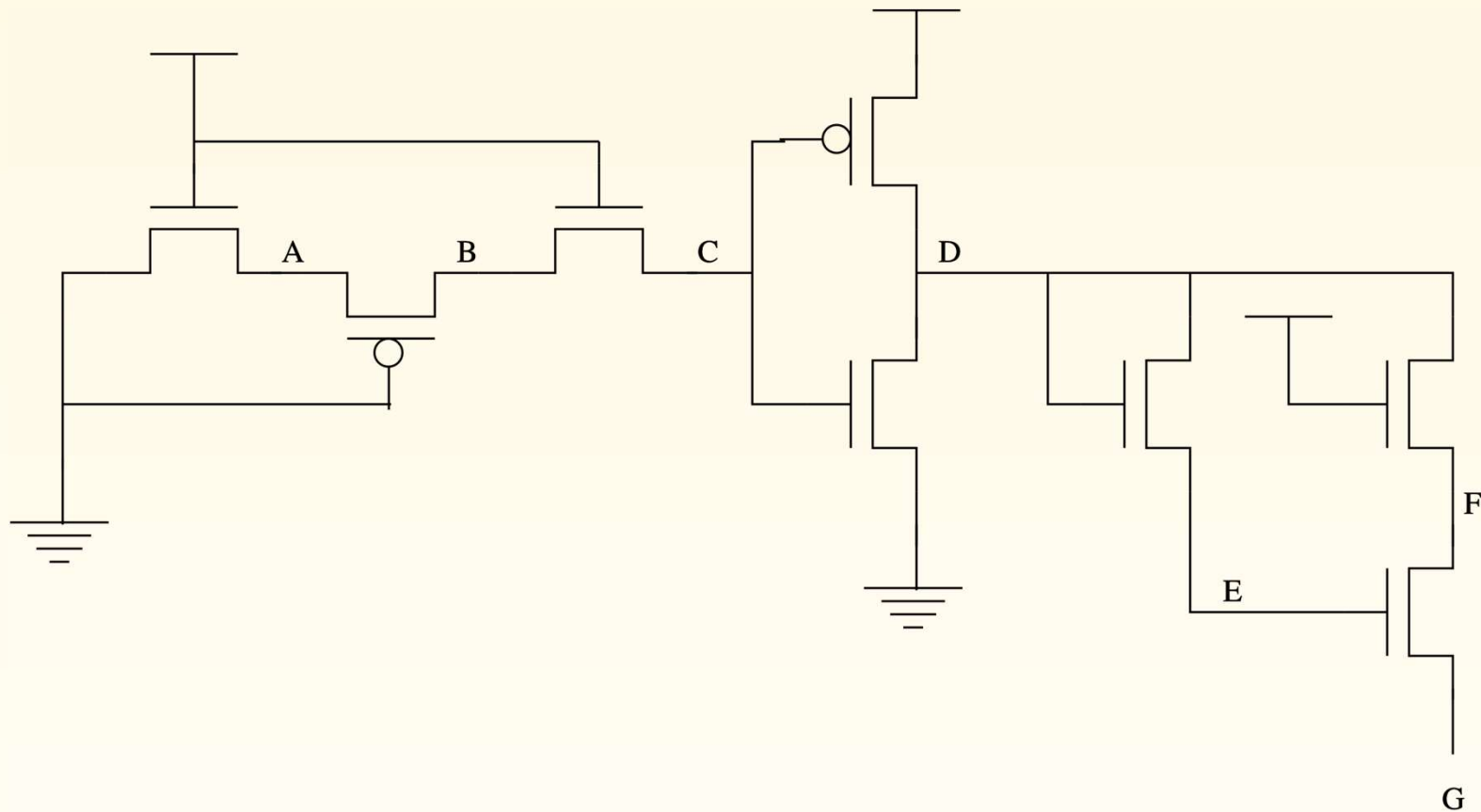
Pass Transistor Ckts



Example 1

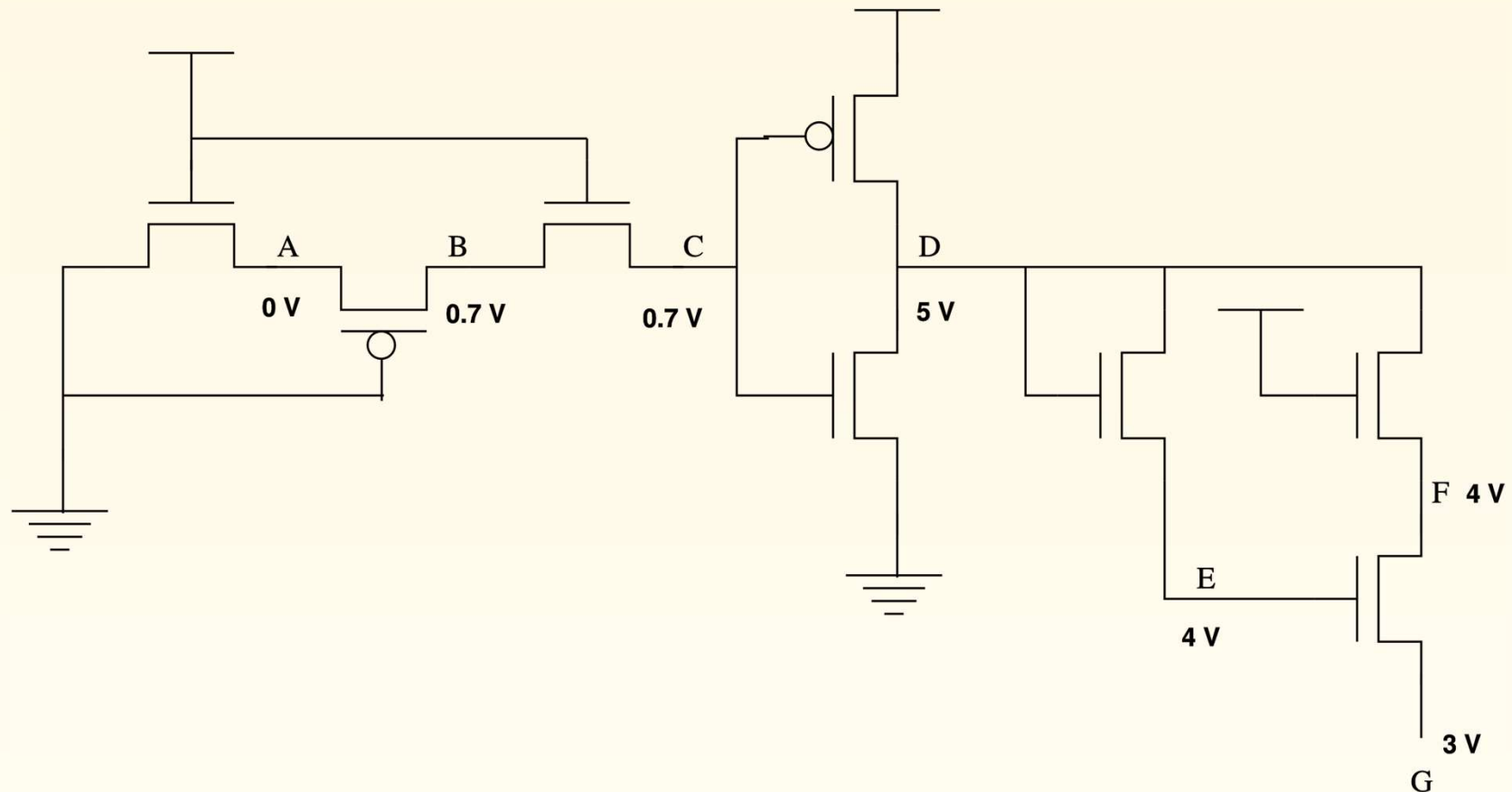
Assumption: initial voltage on each node is 2.5 volts

Relevant transistor parameters are, $V_{dd} = 5V$, $V_{tn} = 1V$ and $|V_{tp}| = 0.7V$



Example 1, Cont'd

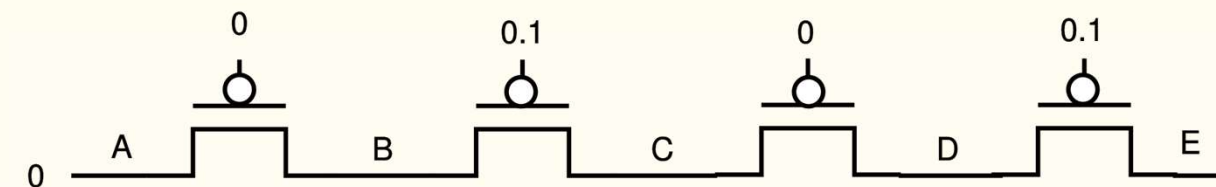
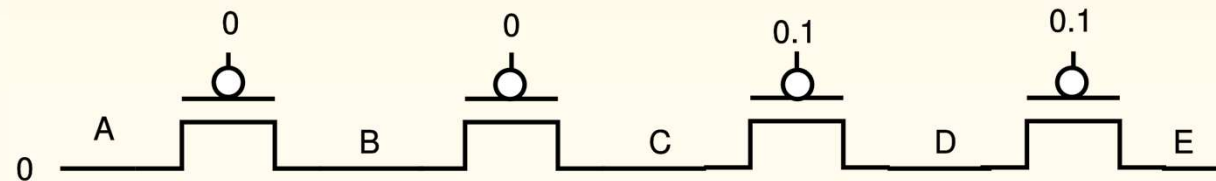
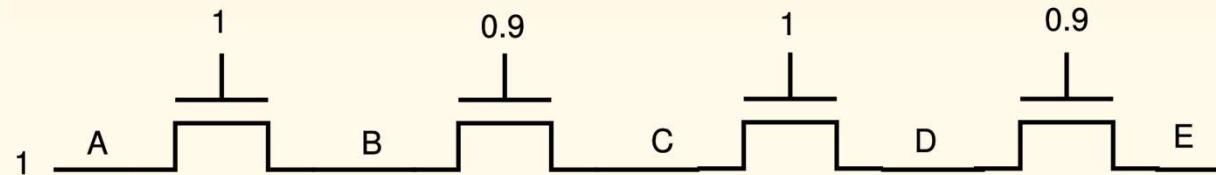
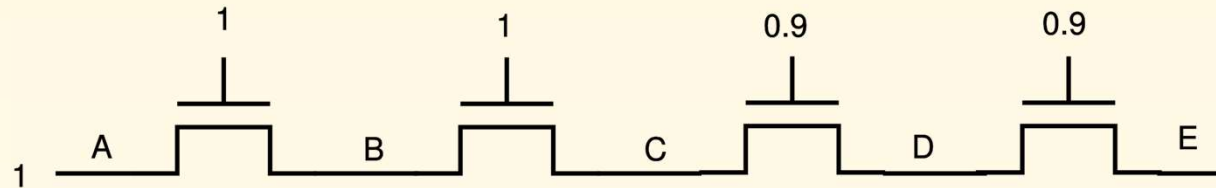
$$V_{dd} = 5V, V_{tn} = 1V \text{ and } |V_{tp}| = 0.7V$$



Example 2

Assume: initial voltage of 0.5V on all the internal nodes

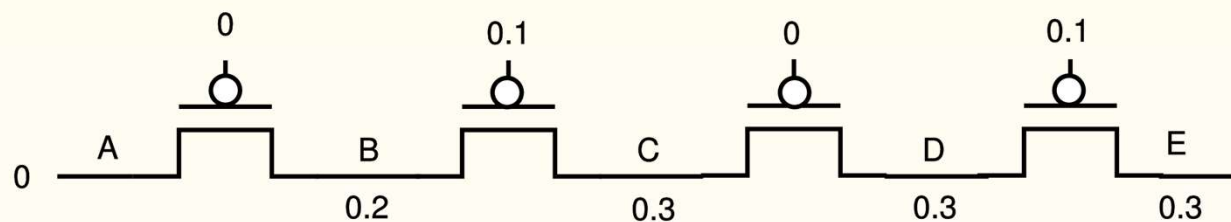
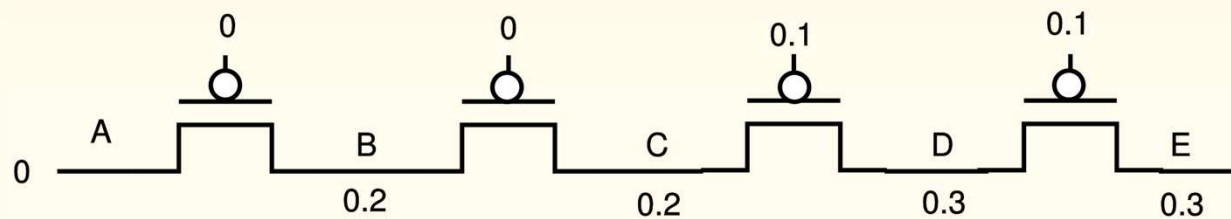
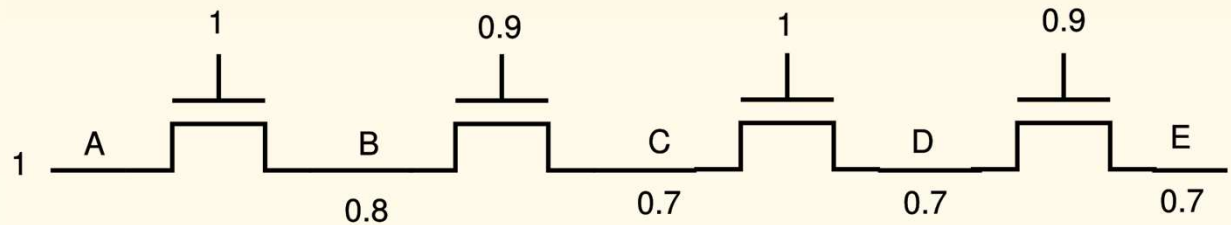
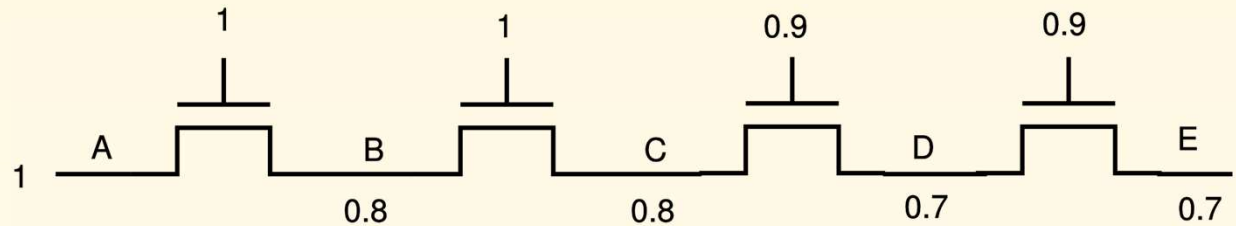
$$V_{dd} = 1.0V, V_{tn} = 0.2V \text{ and } |V_{tp}| = 0.2V$$



Example 2, Cont'd

Assume: initial voltage of 0.5V on all the internal nodes

$$V_{dd} = 1.0V, V_{tn} = 0.2V \text{ and } |V_{tp}| = 0.2V$$



Transmission Gates

- Pass transistors produce degraded outputs
- Transmission gates pass both 0 and 1 well

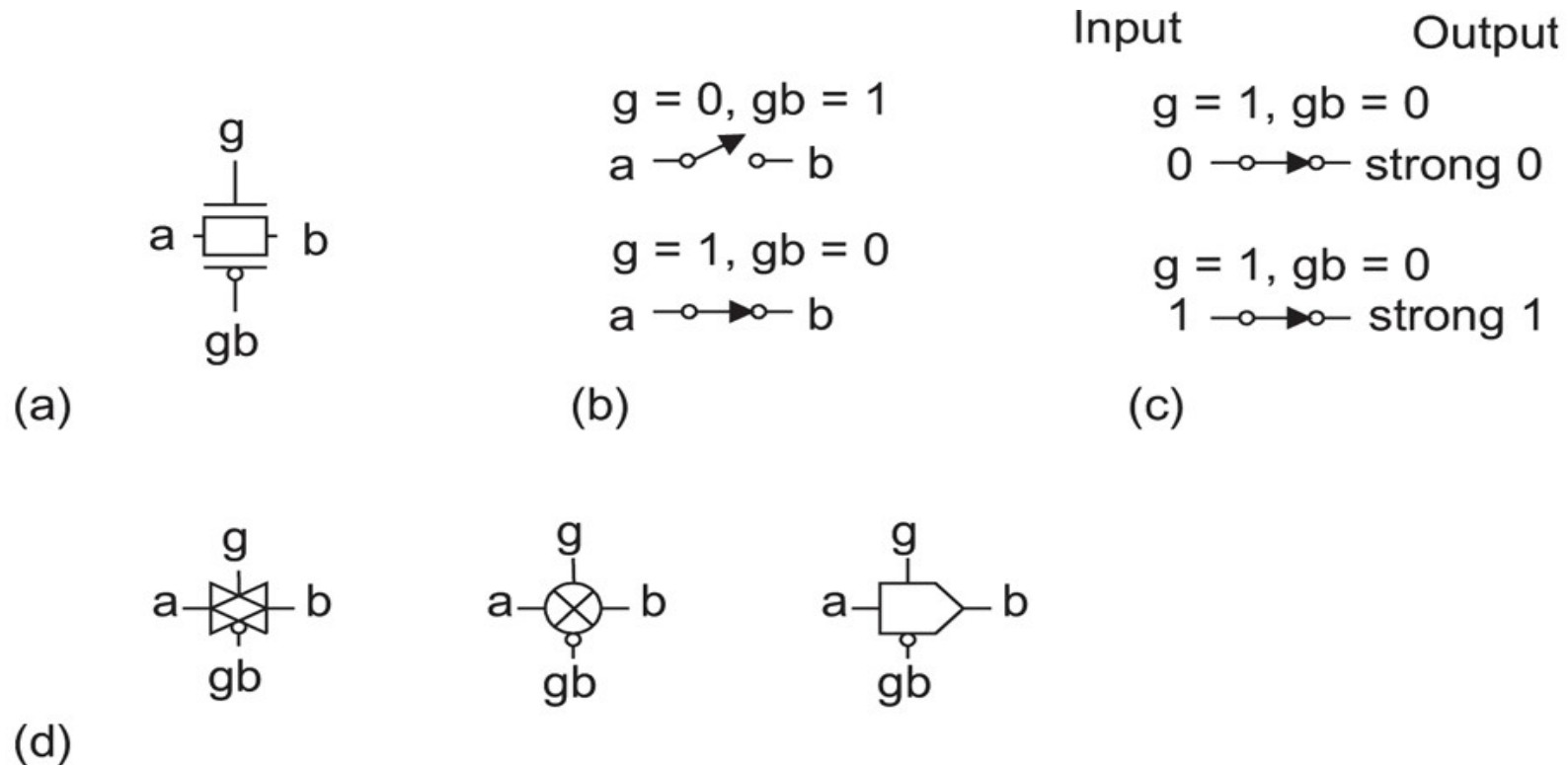
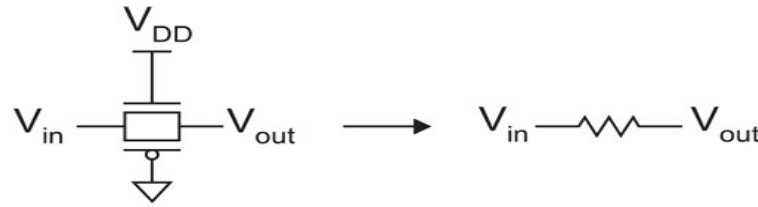


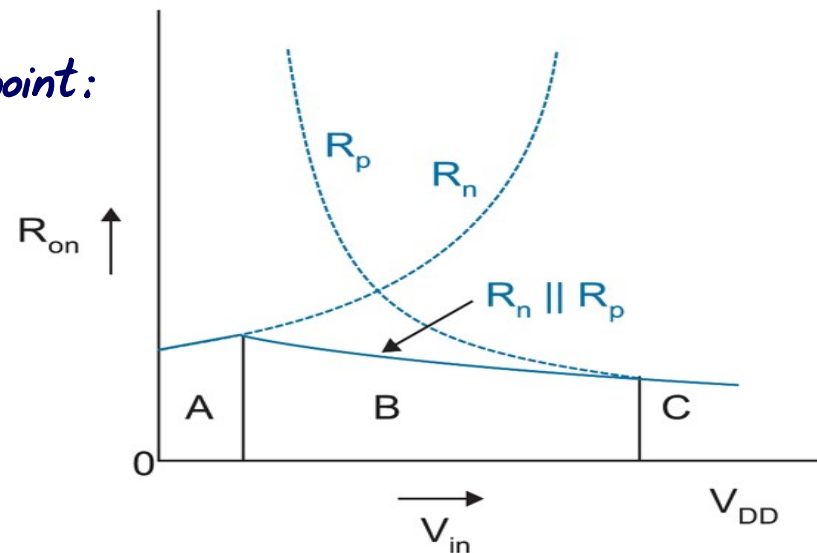
FIG 1.20 Transmission gate

Transmission gate ON resistance



At a given operating point:

$$R = \left(\frac{\partial I_{ds}}{\partial V_{ds}} \right)^{-1}$$



Input voltage V_{in} is swept from GND to V_{DD}

FIG 2.32 Resistance of a transmission gate as a function of input voltage

Review

1. What are conditions for V_{gs} when the nMOS is in cutoff, linear, and saturated modes?
2. What is noise margin?
3. What is transmission gate and its applications?
4. What are V_{tp} and V_{tn} ?
5. Give expressions for the output voltage.

